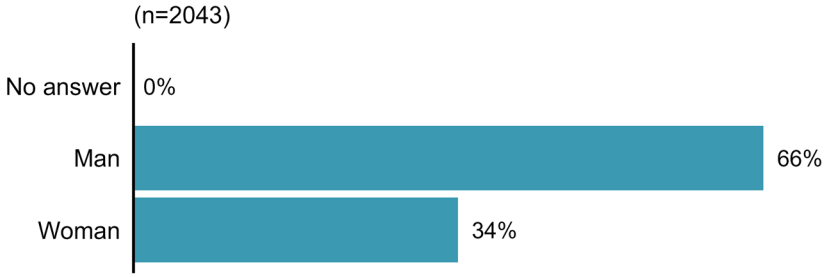
Survey 3 – Experiment participants

The survey, conducted among 2,043 households as part of a pricing experiment, is detailed in the following paper: *Hofmann, Matthias, and Turid Siebenbrunner. 'A Rich Dataset of Hourly Residential Electricity Consumption Data and Survey Answers from the iFlex Dynamic Pricing Experiment'. Data in Brief 50, No. 109571 (1 October 2023). https://doi.org/10.1016/j.dib.2023.109571*.

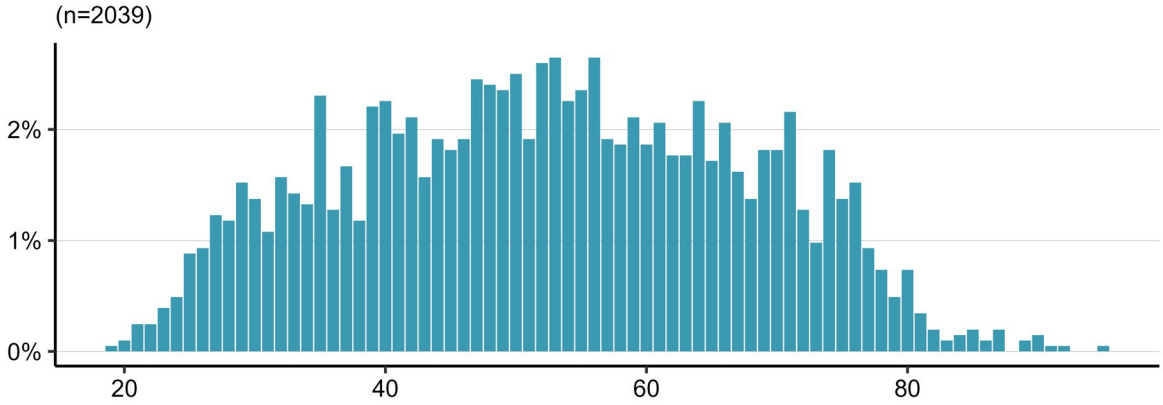
The survey data are published at Zenodo: https://zenodo.org/records/8248802

This document is published at Zenodo: https://zenodo.org/records/11580541

Question QI1: Sex



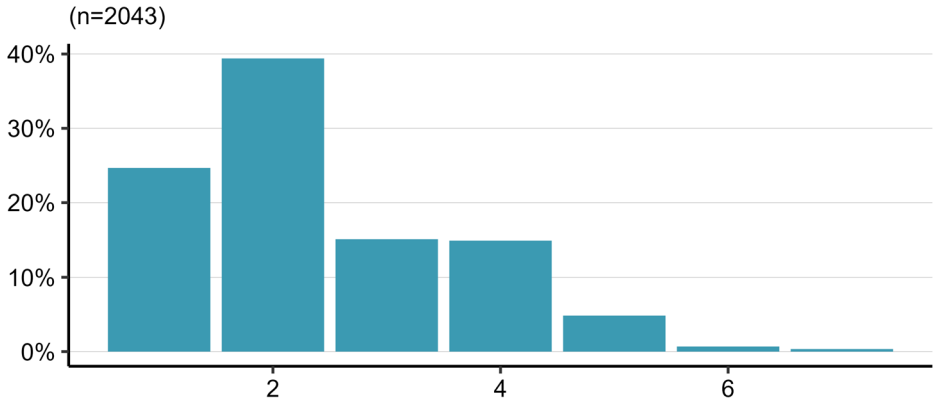
Question QI2: Age



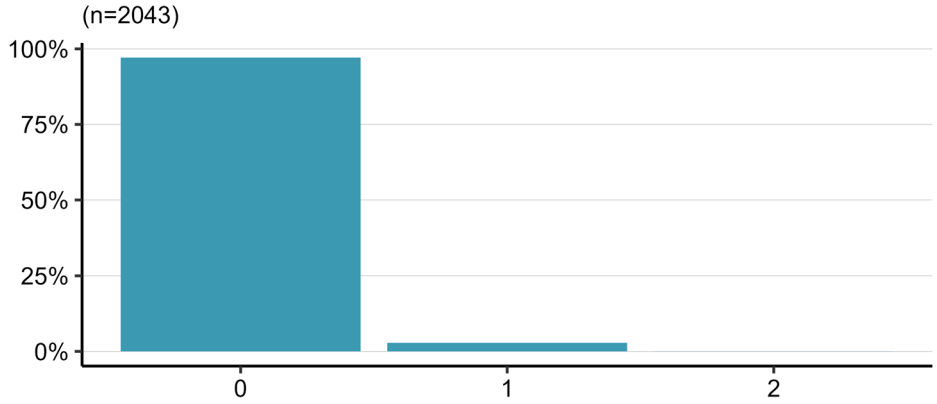
Question QI5: Are you responsible for paying the electricity bill?



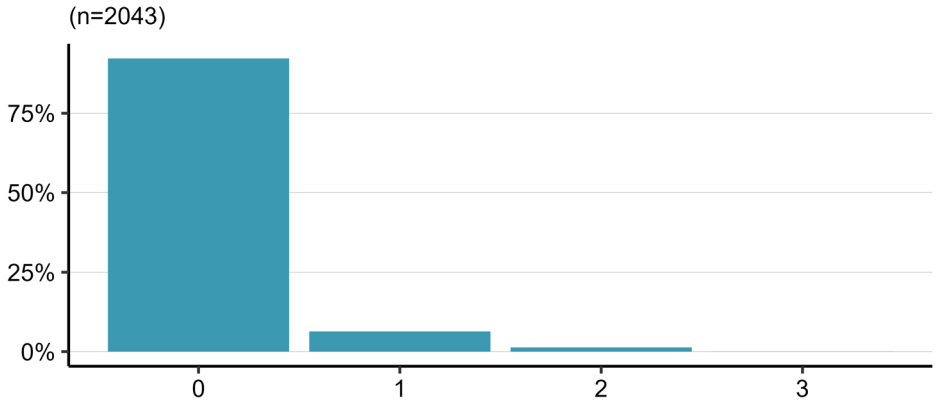
Question QI6: How many people, yourself included, usually lived in your household in the period January - March 2021?



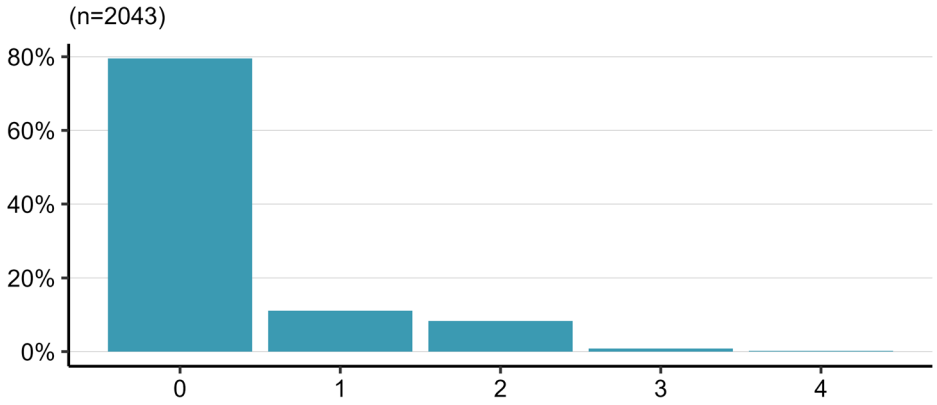
Question QI7_1: In what age group (s) are the household residents? How many of the household's residents are… 0-1 years



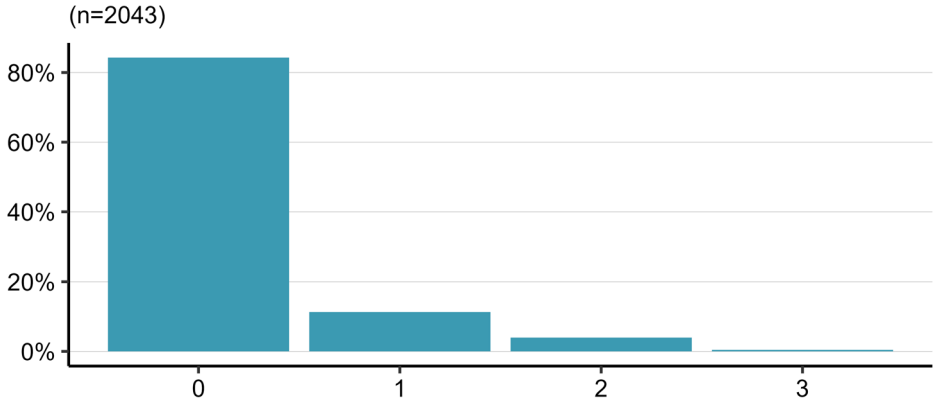
Question QI7_2: In what age group (s) are the household residents? How many of the household's residents are… 2-5 years



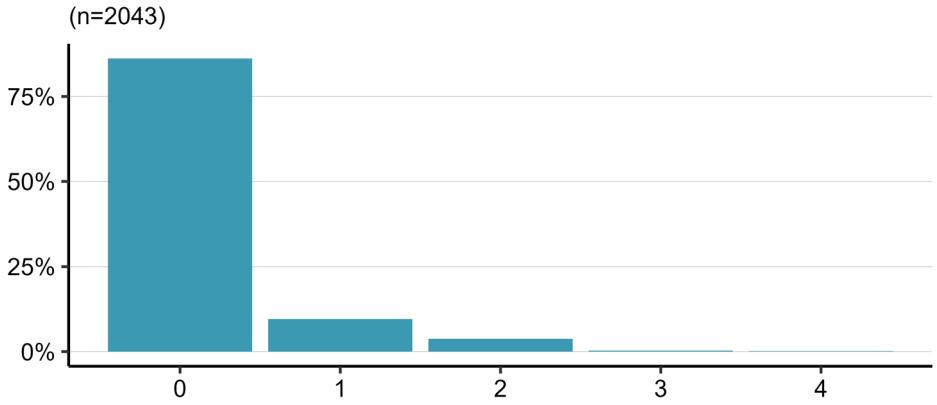
Question QI7_3: In what age group (s) are the household residents? How many of the household's residents are… 6-15 years



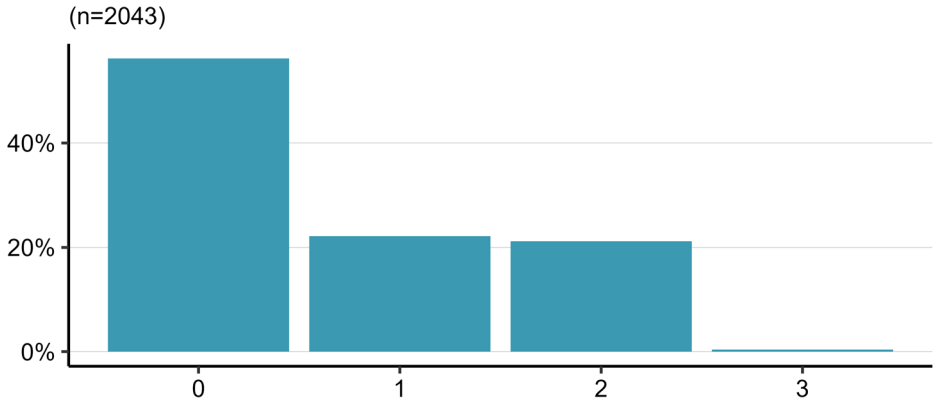
Question QI7_4: In what age group (s) are the household residents? How many of the household's residents are… 16-22 years



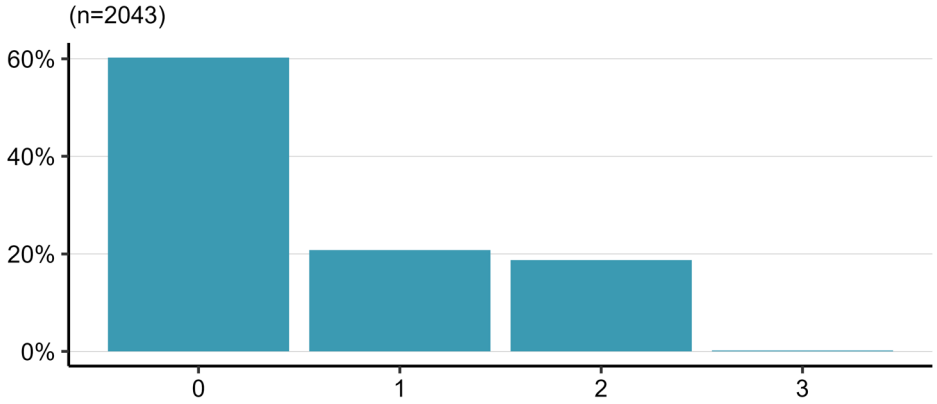
Question QI7_5: In what age group (s) are the household residents? How many of the household's residents are… 23-29 years



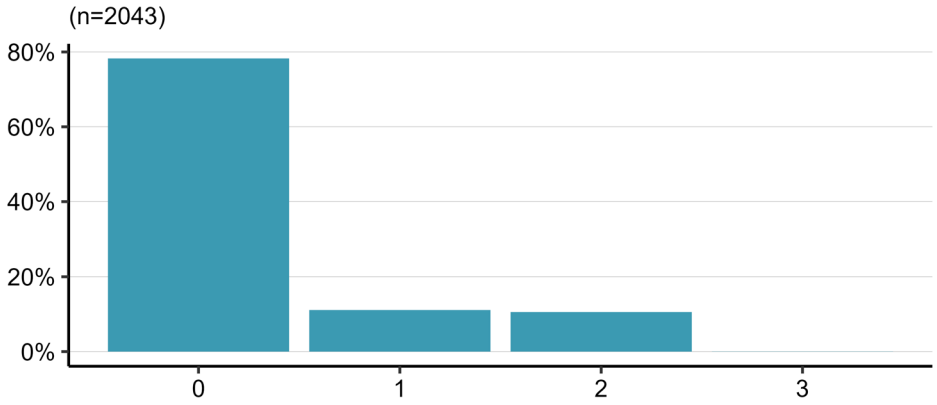
Question QI7_6: In what age group (s) are the household residents? How many of the household's residents are… 30-49 years



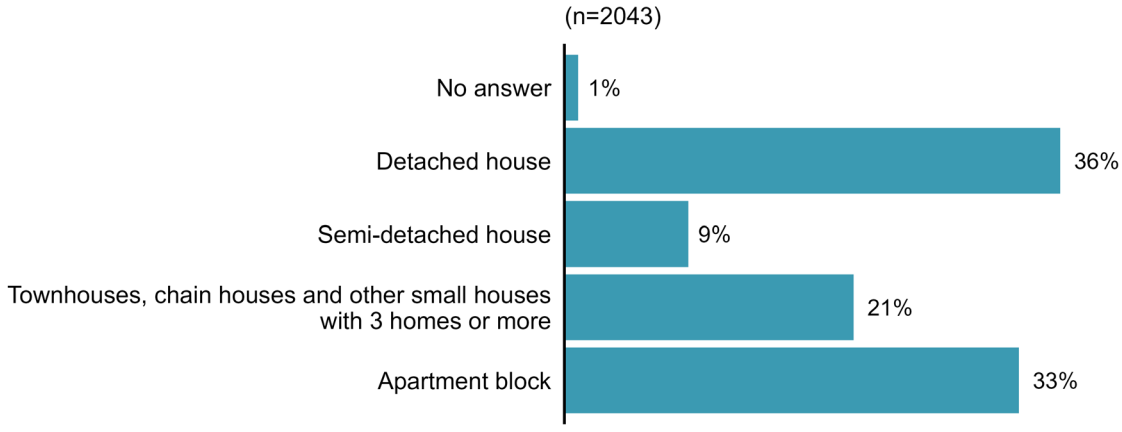
Question QI7_7: In what age group (s) are the household residents? How many of the household's residents are… 50-66 years



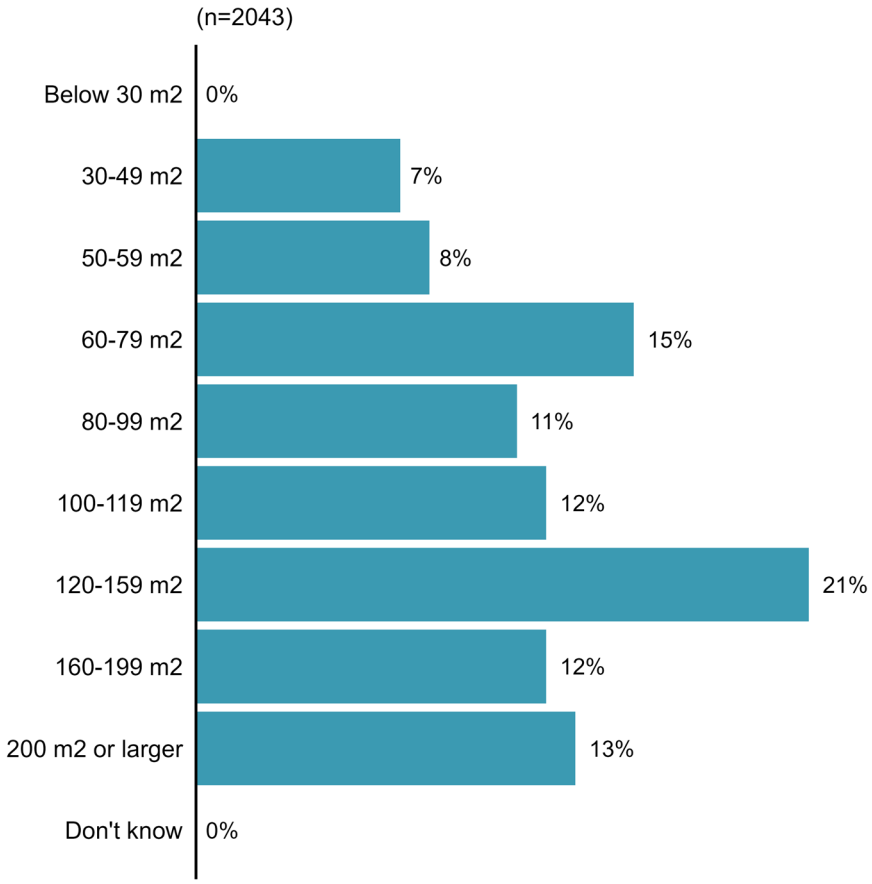
Question QI7_8: In what age group (s) are the household residents? How many of the household's residents are…67 years and older



Question Q1: What type of building do you live in? Is it a…



Question Q2: How large is the home in square meters?



Question Q3: Do you own the home?



(IF Question Q3 = Yes)
Question Q4: Do you have a rental unit in the home?

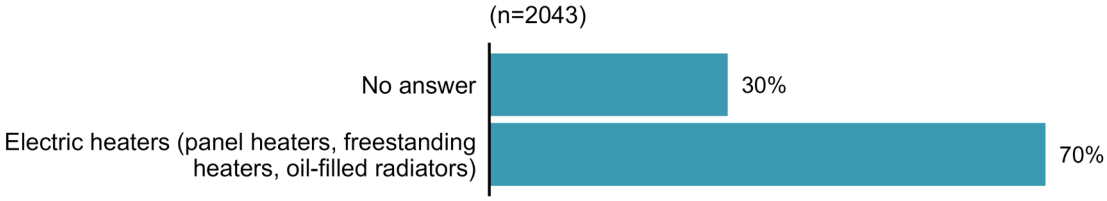


(IF Question Q4 = Yes)

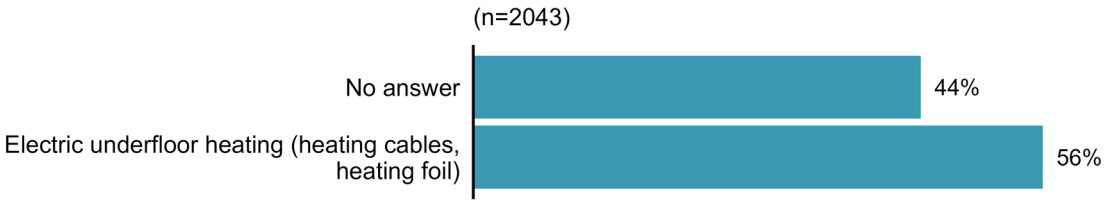
Question Q5: Does the rental unit has a separate electricity meter?



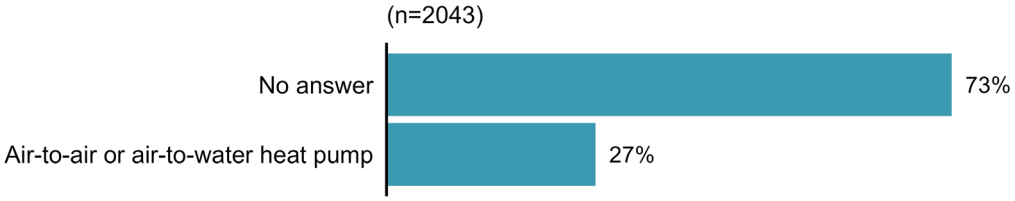
Question Q6_1: How is the home heated? Electric heaters (panel heaters, freestanding heaters, oil-filled radiators)



Question Q6_2: How is the home heated? Electric underfloor heating (heating cables, heating foil)



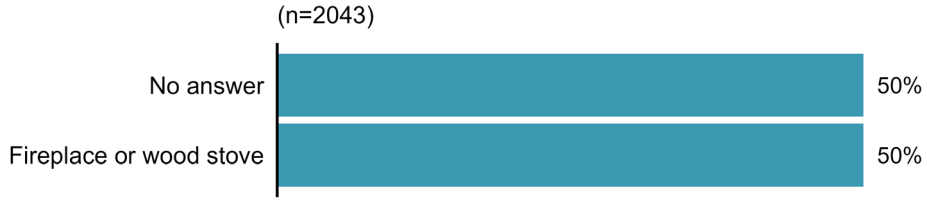
Question Q6_3: How is the home heated? Air-to-air or air-to-water heat pump



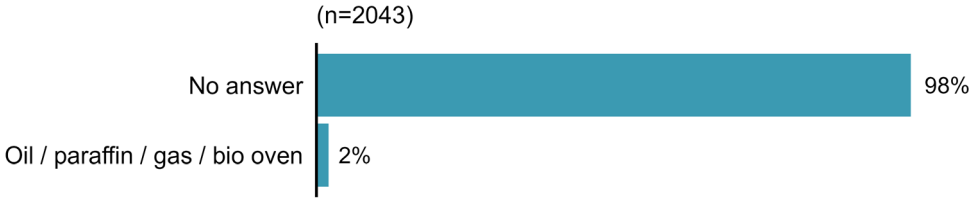
Question Q6_4: How is the home heated? Geothermal / rock heat



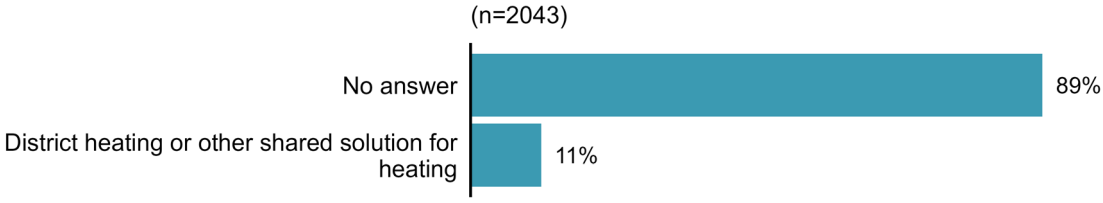
Question Q6_5: How is the home heated? Fireplace or wood stove



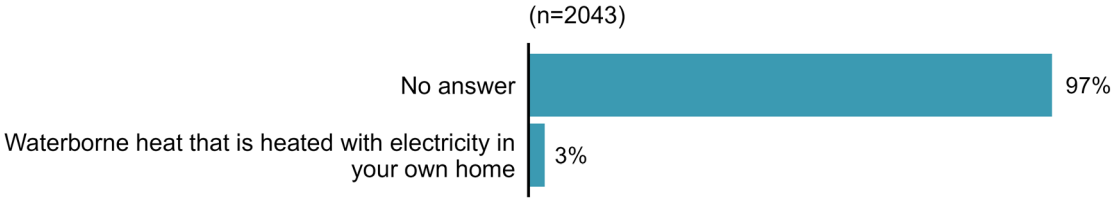
Question Q6_6: How is the home heated? Oil / paraffin / gas / bio oven



Question Q6_7: How is the home heated? District heating or other shared solution for heating in the building



Question Q6_8: How is the home heated? Waterborne heat that is heated with electricity in your own home



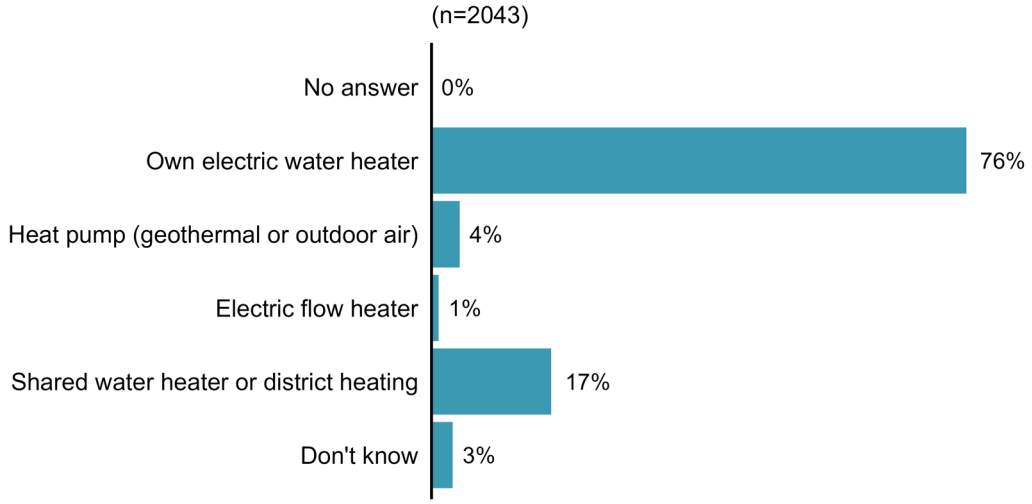
Question Q6_9: How is the home heated? Other type, enter:



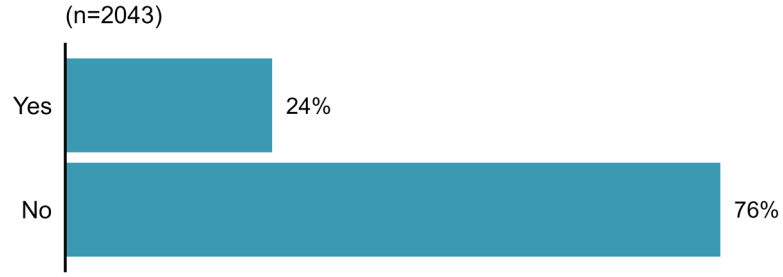
Question Q6_10: How is the home heated? Don't know



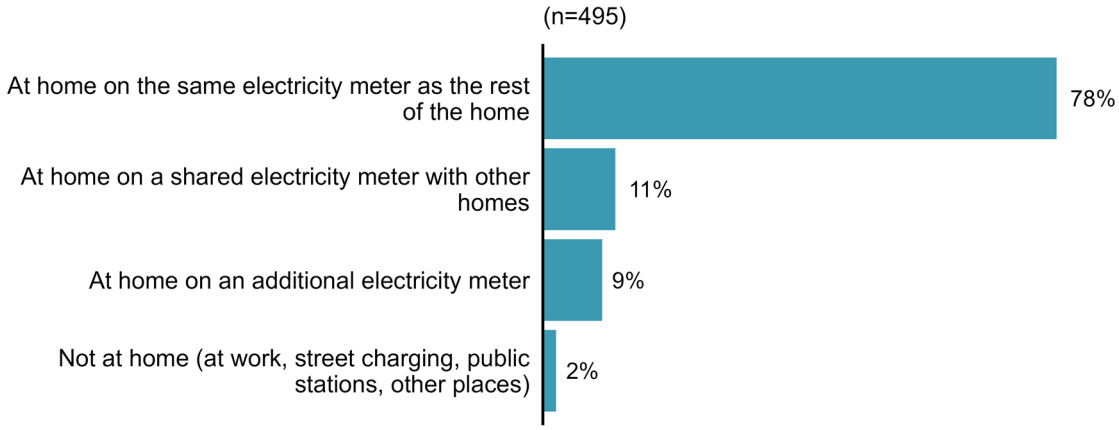
Question Q7: How is the hot water heated in your home?



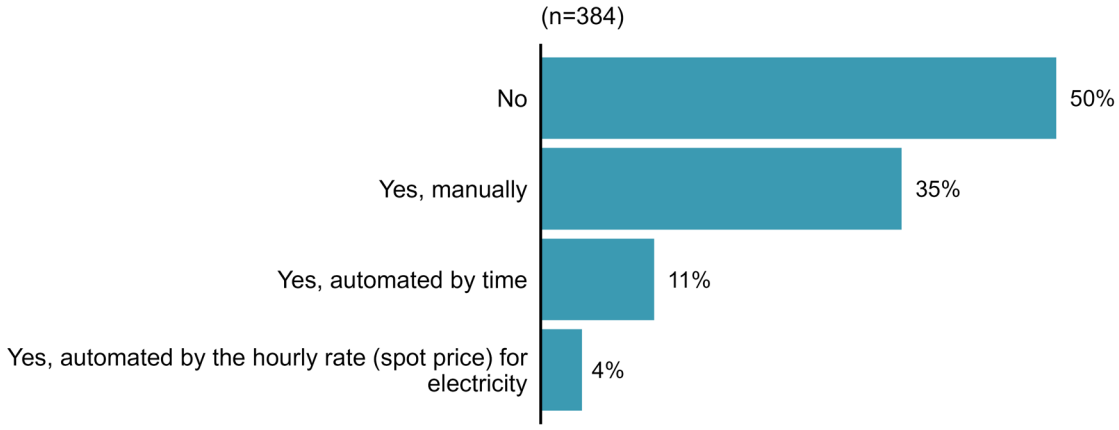
Question Q8: Do you have an electric car(s) in the household that is occasionally charged at home?



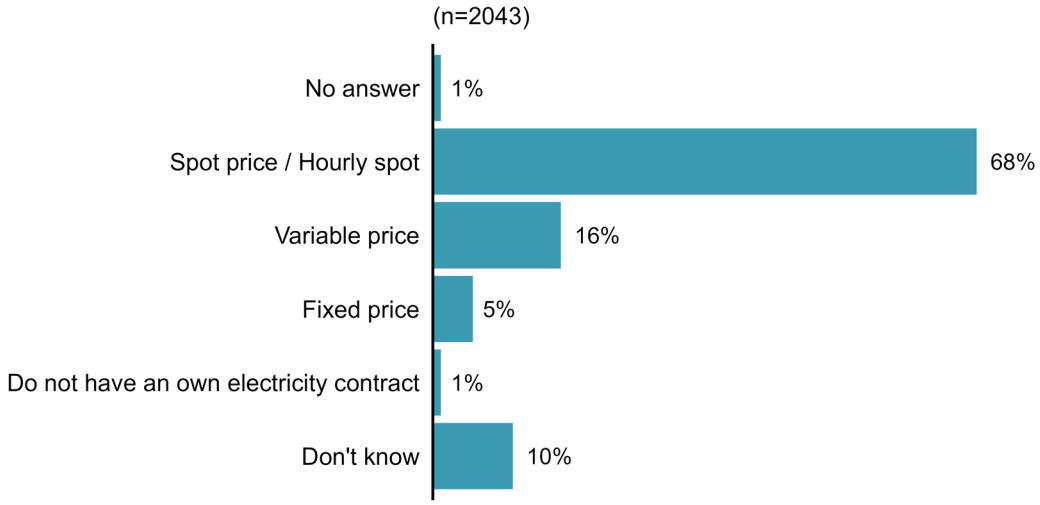
(IF Question Q8 = Yes)
Question Q9: How is the electric car(s) usually charged?



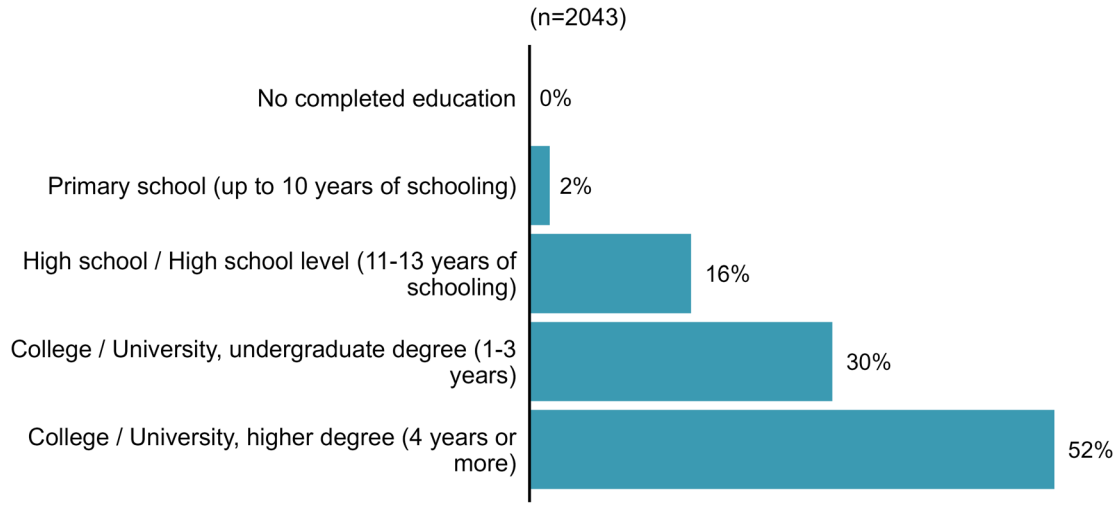
(IF Question Q9 = At home on the same electricity meter as the rest of the home)
Question Q10: Do you control the charging of the electric car(s) to avoid hours of high electricity prices?



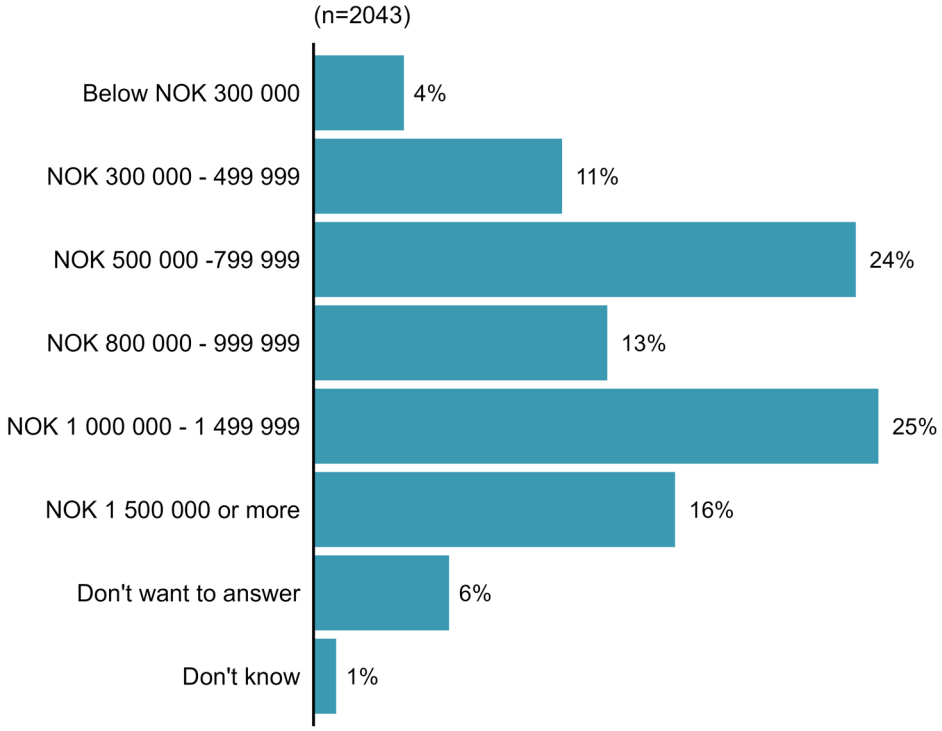
Question Q11: What type of electricity contract do you have?



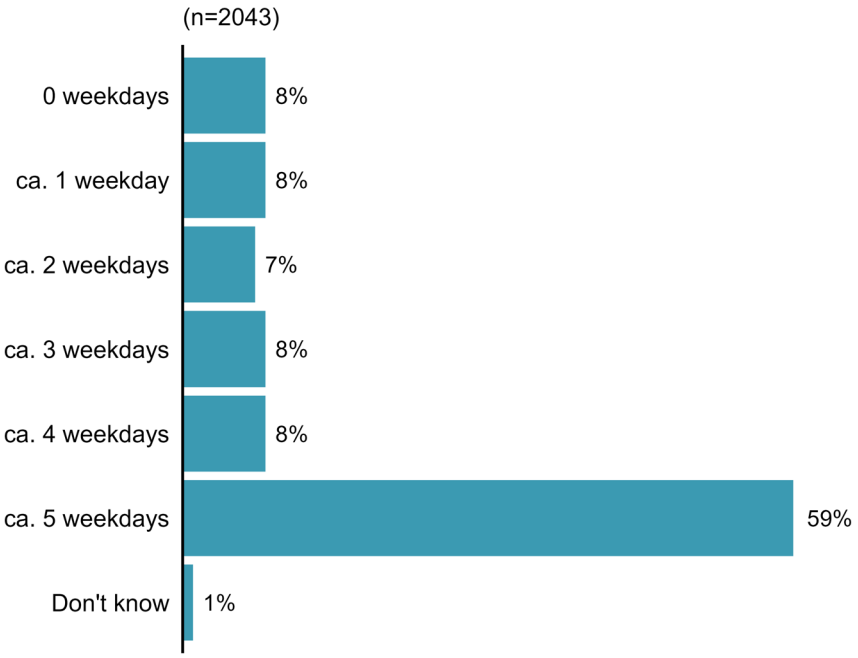
Question Q12: What is the highest completed education in the household?



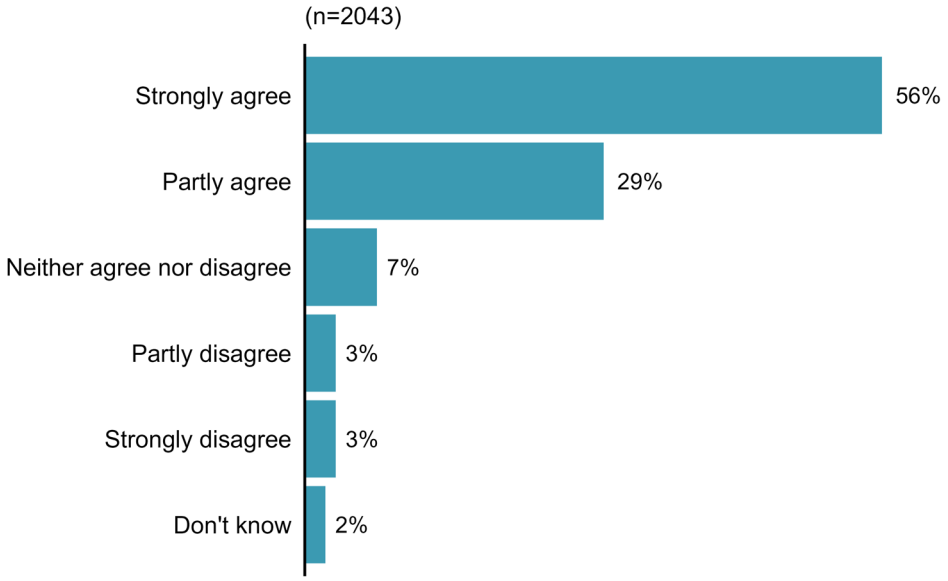
Question Q13: What is the total gross income before taxes of the household?



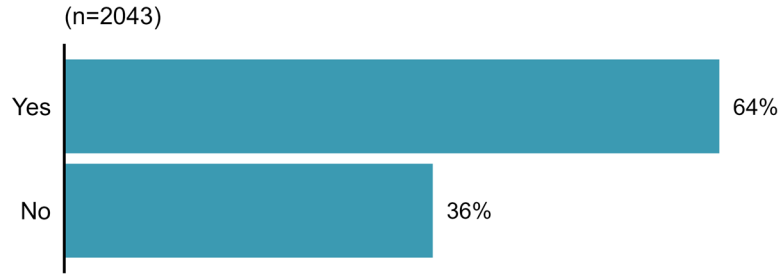
Question Q14: On how many weekdays (Monday-Friday) was at least one person at home during the day (9 am - 4 pm) during an average working week in the period January - March 2021?



Question Q15: Do you agree or disagree with the following statements? People who adjust their electricity consumption based on the price of electricity should be able to save on electricity costs.



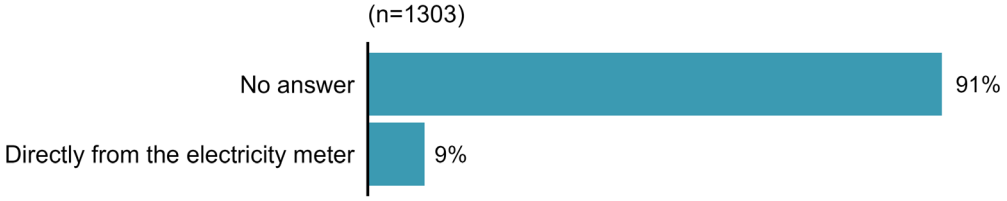
Question Q16: Do you usually follow actively your own electricity consumption?



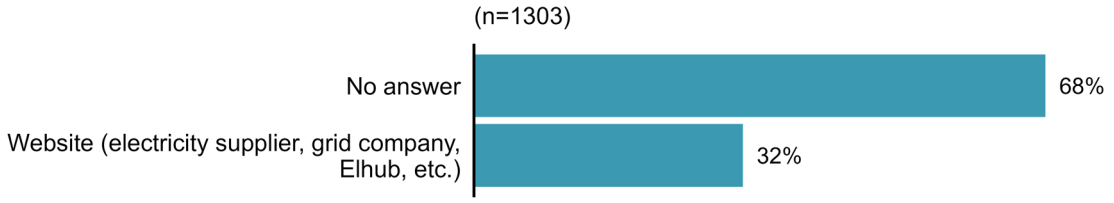
(IF Question Q16 = Yes)
Question Q17_1: How do you get information about your electricity consumption? Electricity bill



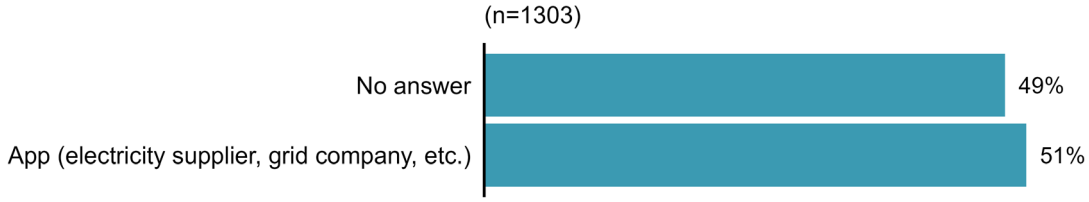
(IF Question Q16 = Yes)
Question Q17_2: How do you get information about your electricity consumption? Directly from the electricity meter



(IF Question Q16 = Yes)
Question Q17_3: How do you get information about your electricity consumption? Website (electricity supplier, grid company, Elhub, etc.)



(IF Question Q16 = Yes)
Question Q17_4: How do you get information about your electricity consumption? App (electricity supplier, grid company, etc.)



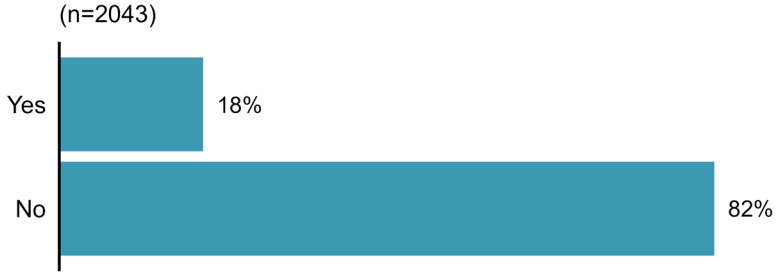
(IF Question Q16 = Yes)
Question Q17_5: How do you get information about your electricity consumption? Email or SMS



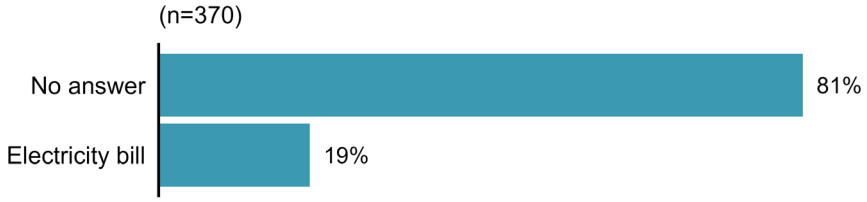
(IF Question Q16 = Yes)
Question Q17_6: How do you get information about your electricity consumption? Other:



Question Q18: Do you follow actively how electricity prices vary from day to day and hour to hour?



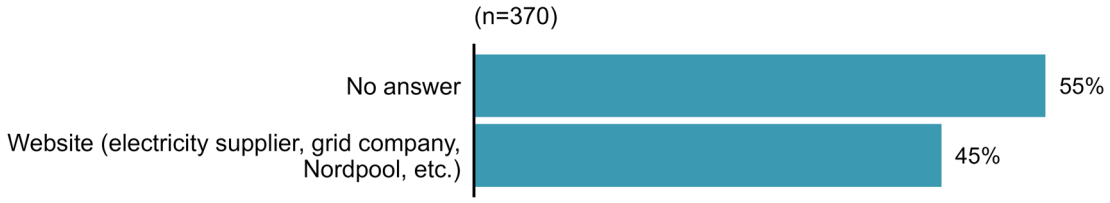
(IF Question Q18 = Yes)
Question Q19_1: How do you get information about electricity prices? Electricity bill



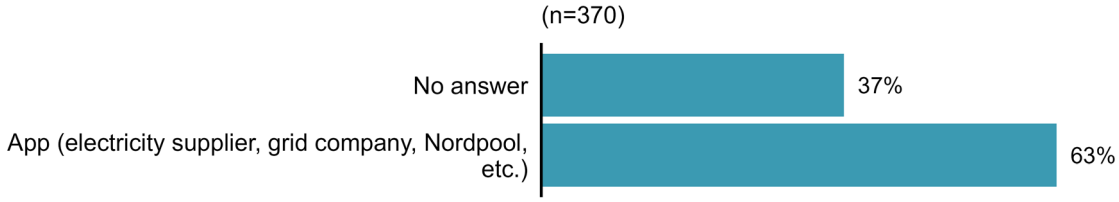
(IF Question Q18 = Yes)
Question Q19_2: How do you get information about electricity prices? Media / news



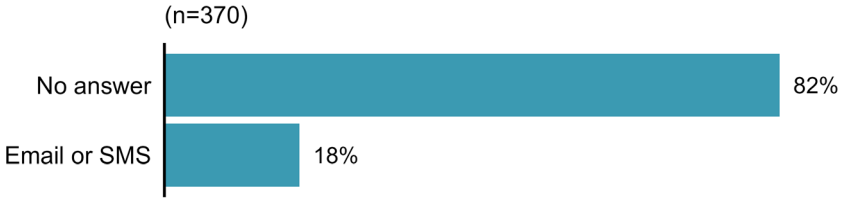
(IF Question Q18 = Yes)
Question Q19_3: How do you get information about electricity prices? Website (electricity supplier, grid company, Nordpool, etc.)



(IF Question Q18 = Yes)
Question Q19_4: How do you get information about electricity prices? App (electricity supplier, grid company, Nordpool, etc.)



(IF Question Q18 = Yes)
Question Q19_5: How do you get information about electricity prices? Email or SMS



(IF Question Q18 = Yes)
Question Q19_6: How do you get information about electricity prices? From work



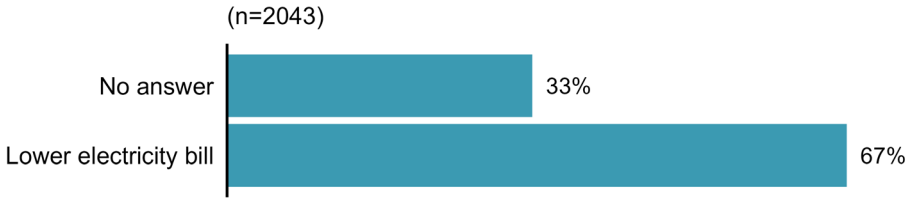
(IF Question Q18 = Yes)
Question Q19_7: How do you get information about electricity prices? Other:



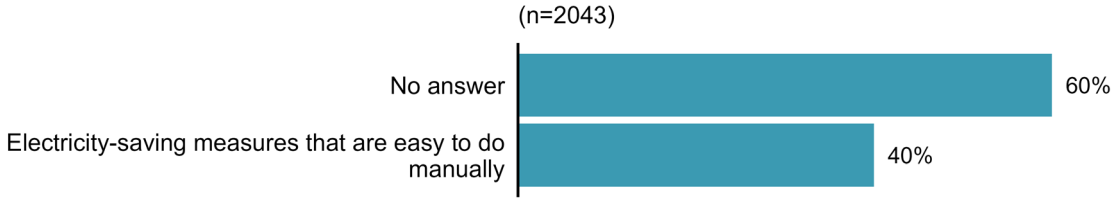
(IF Question Q18 = Yes)
Question Q19_8: How do you get information about electricity prices? Don't know



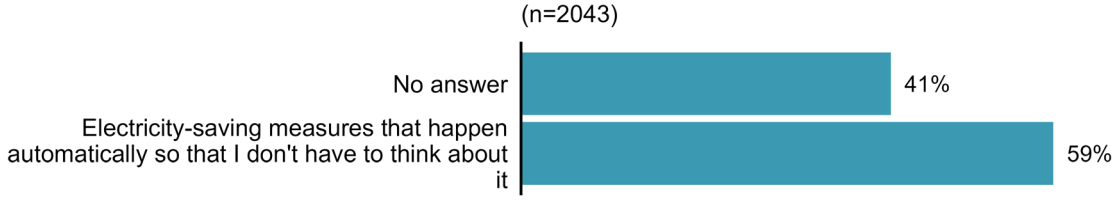
Question Q20_1: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? Lower electricity bill



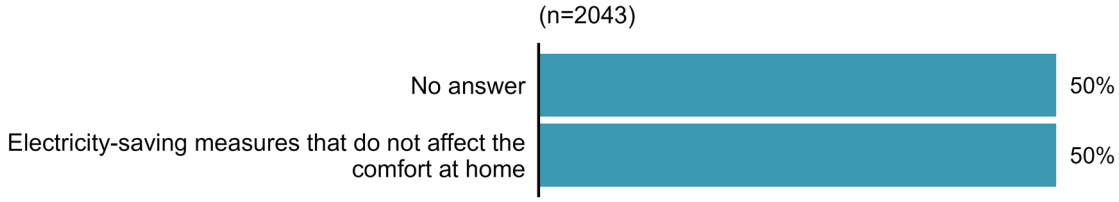
Question Q20_2: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? Electricity-saving measures that are easy to do manually



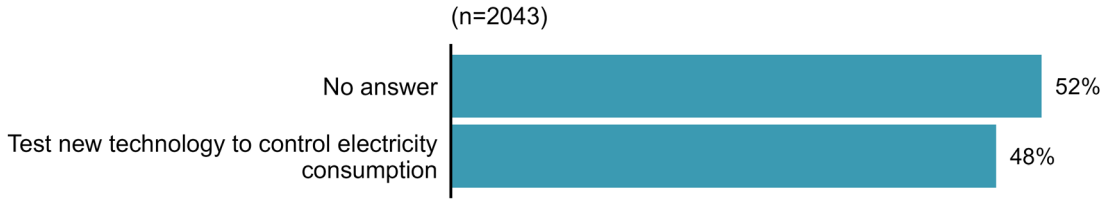
Question Q20_3: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? Electricity-saving measures that happen automatically so that I don't have to think about it



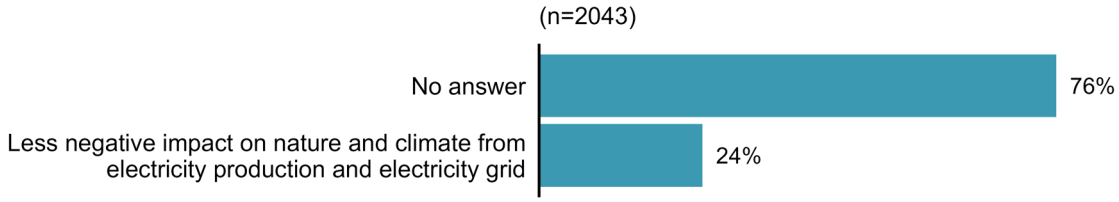
Question Q20_4: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? Electricity-saving measures that do not affect the comfort at home



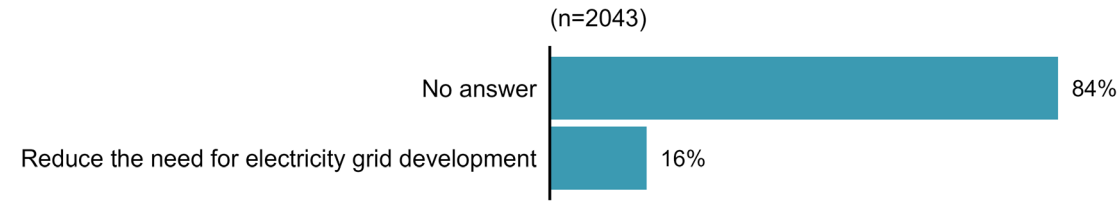
Question Q20_5: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? Test new technology to control electricity consumption



Question Q20_6: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? Less negative impact on nature and climate from electricity production and electricity grid



Question Q20_7: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? Reduce the need for electricity grid development



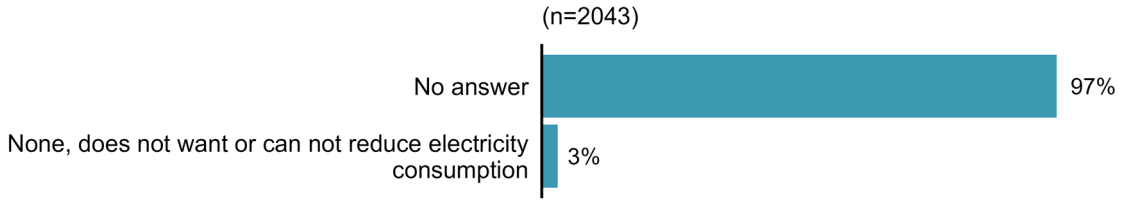
Question Q20_8: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? If many others do



Question Q20_9: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? Other:



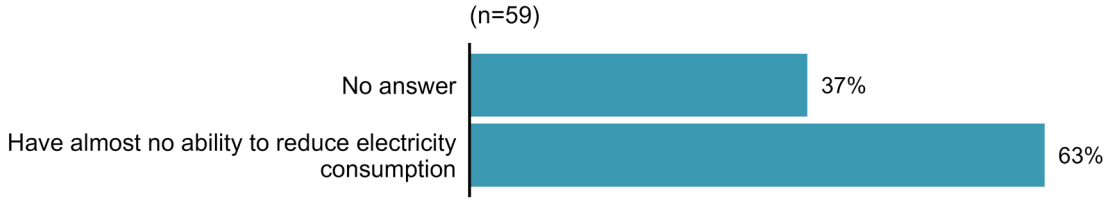
Question Q20_11: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? None, does not want or can not reduce electricity consumption



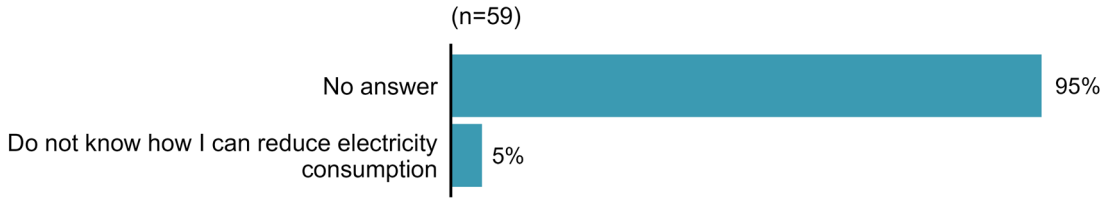
Question Q20_12: Could any of the following motivate you to reduce your electricity consumption in hours of high electricity prices, when it is cold outside and you are at home? Don't know



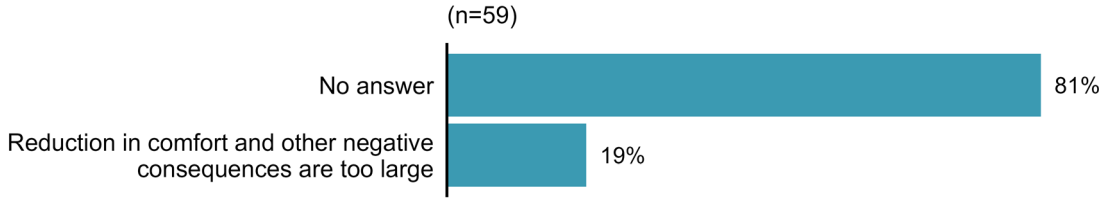
(IF Question Q20_11 = None, does not want or can not reduce electricity consumption)
Question Q21_1: Why do you not want to reduce your electricity consumption? Have almost no ability to reduce electricity consumption



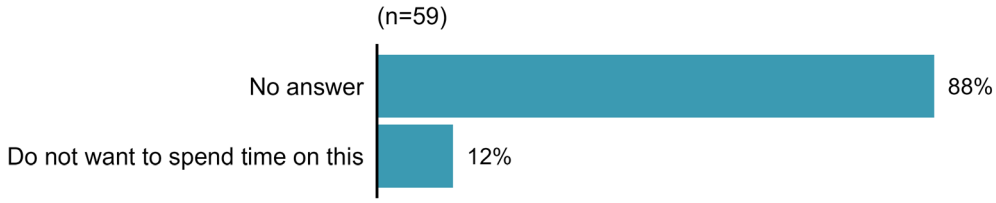
(IF Question Q20_11 = None, does not want or can not reduce electricity consumption)
Question Q21_2: Why do you not want to reduce your electricity consumption? Do not know how I can reduce electricity consumption



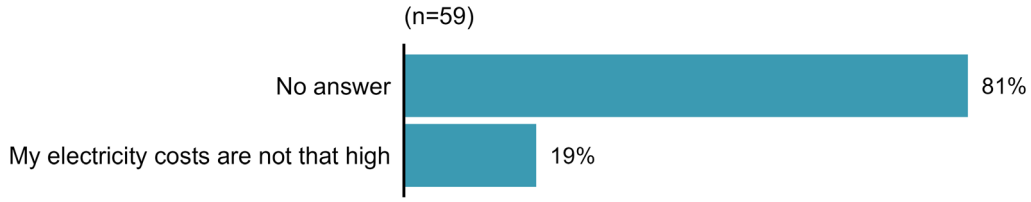
(IF Question Q20_11 = None, does not want or can not reduce electricity consumption)
Question Q21_3: Why do you not want to reduce your electricity consumption? Reduction in comfort and other negative consequences are too large



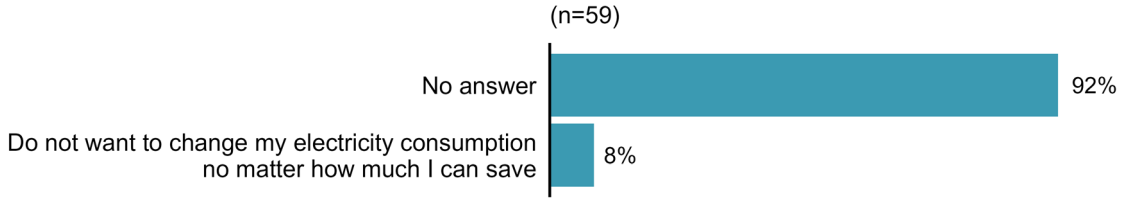
(IF Question Q20_11 = None, does not want or can not reduce electricity consumption)
Question Q21_4: Why do you not want to reduce your electricity consumption? Do not want to spend time on this



(IF Question Q20_11 = None, does not want or can not reduce electricity consumption)
Question Q21_5: Why do you not want to reduce your electricity consumption? My electricity costs are not that high



(IF Question Q20_11 = None, does not want or can not reduce electricity consumption)
Question Q21_6: Why do you not want to reduce your electricity consumption? Do not want to change my electricity consumption no matter how much I can save



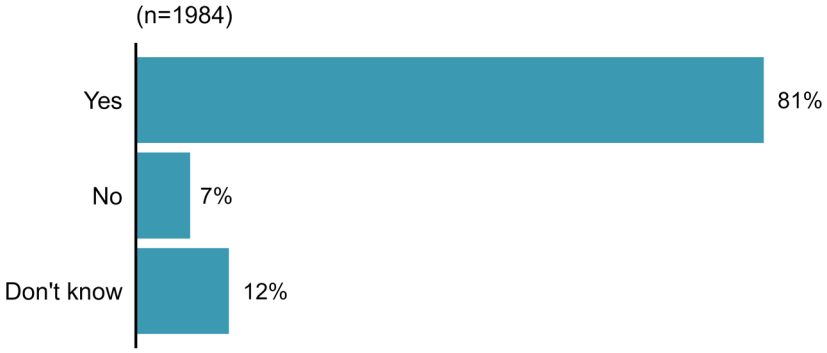
Question Q21_7: Why do you not want to reduce your electricity consumption? Other:



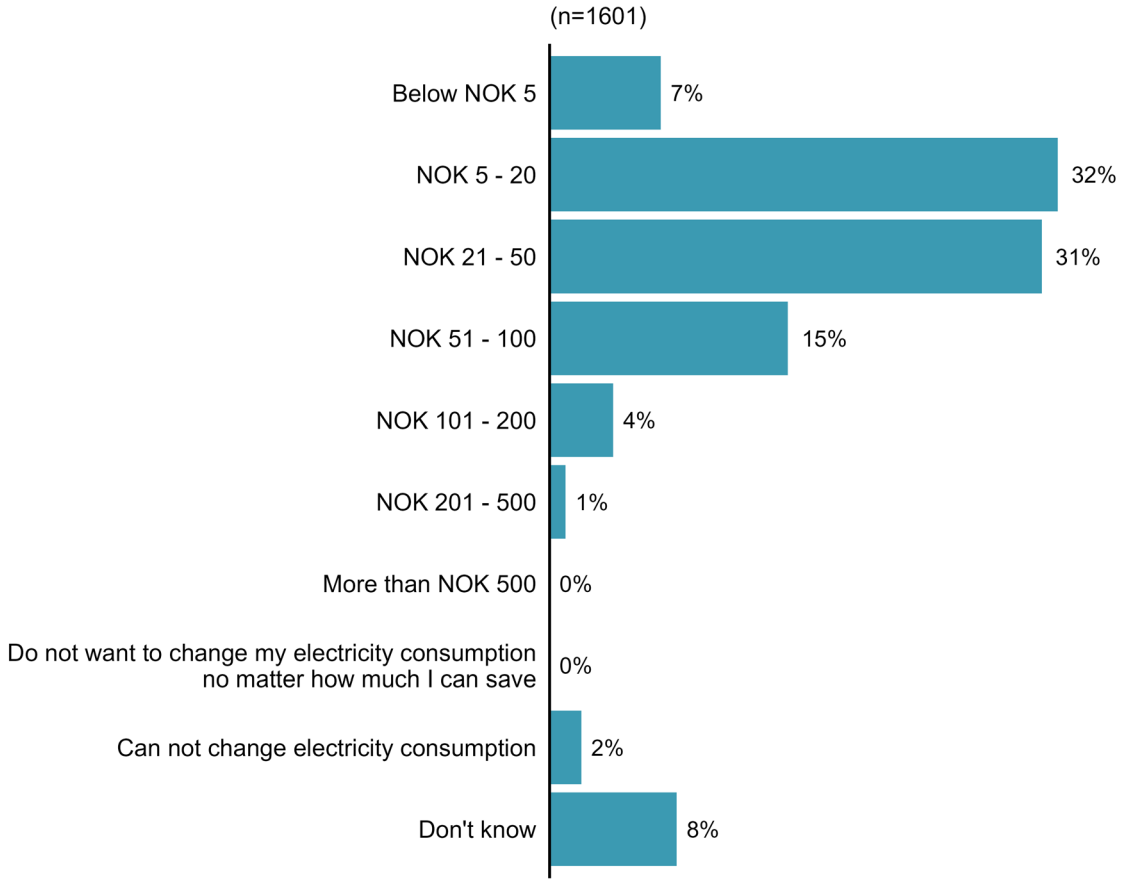
(IF Question Q20_11 = None, does not want or can not reduce electricity consumption)
Question Q21_8: Why do you not want to reduce your electricity consumption? Don't know



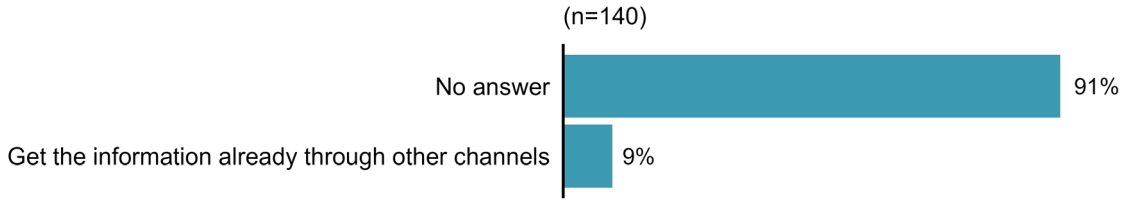
Question Q22: Would you use a free information service that notifies you when the price of electricity the next day would be very high for a few hours?



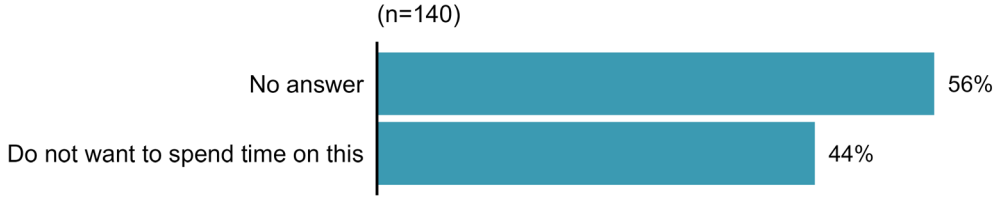
(IF Question Q22 = Yes)
Question Q23: How much higher did the expected cost of electricity for the entire next day have to be for you to want to be notified? … Kroner higher compared to a normal winter day.



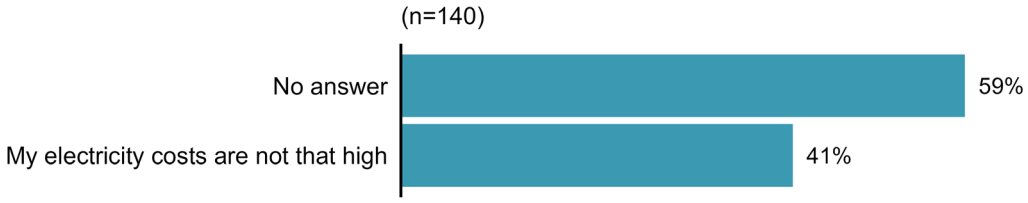
(IF Question Q22 = No)
Question Q24_1: Why do you not want to use such an information service? Get the information already through other channels



(IF Question Q22 = No)
Question Q24_2: Why do you not want to use such an information service? Do not want to spend time on this



(IF Question Q22 = No)
Question Q24_3: Why do you not want to use such an information service? My electricity costs are not that high



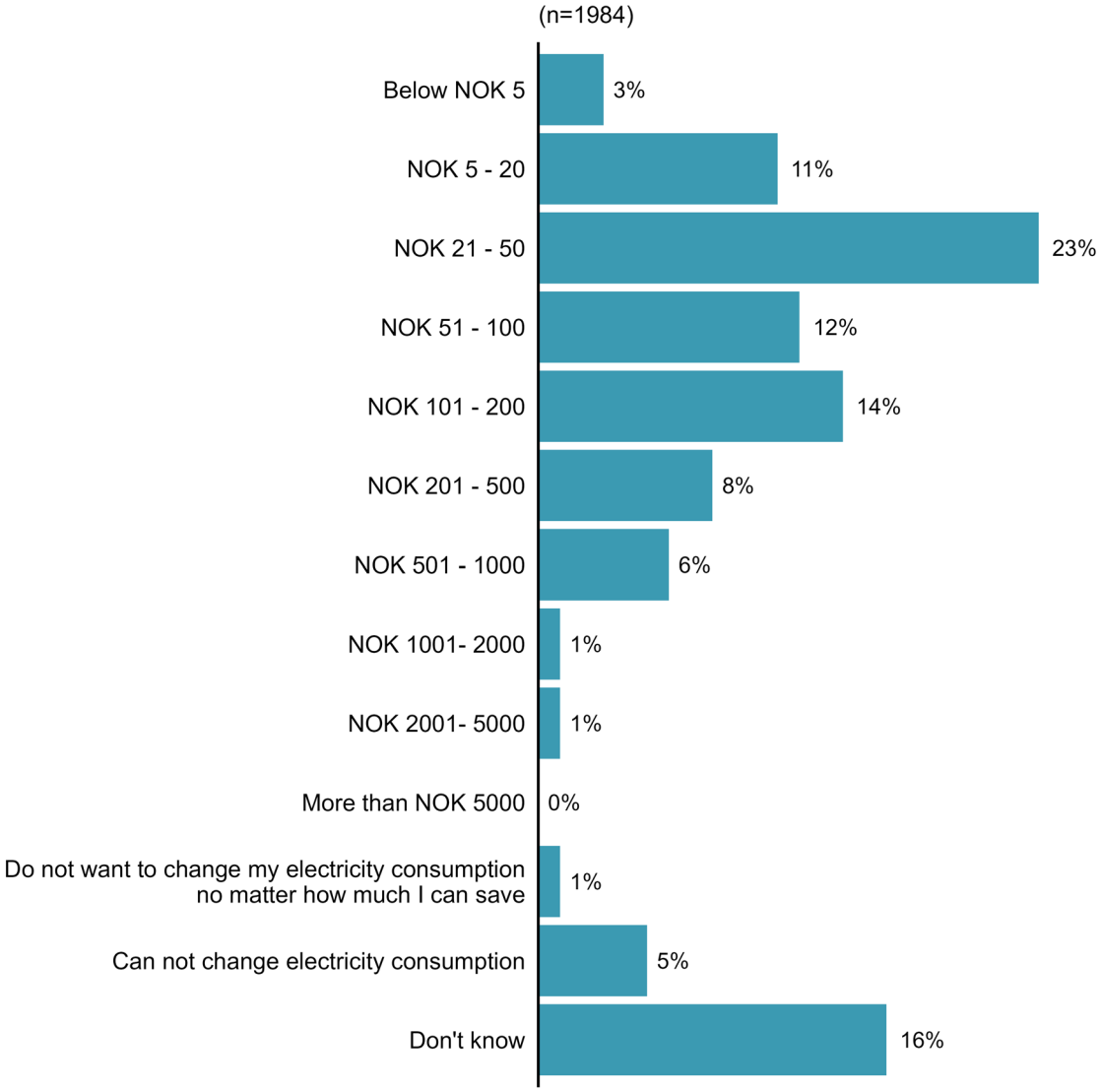
(IF Question Q22 = No)
Question Q24_4: Why do you not want to use such an information service? Other:



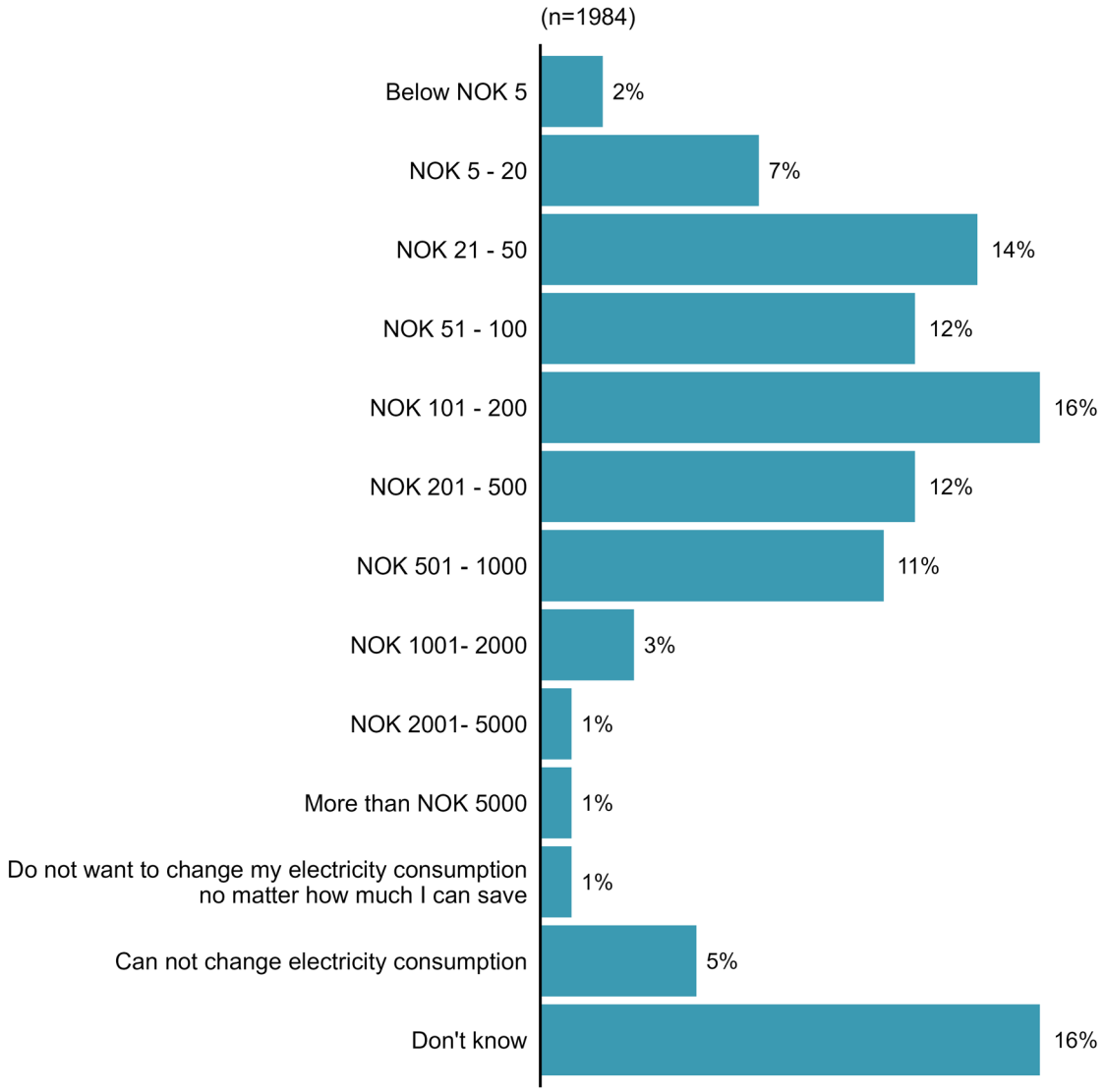
(IF Question Q22 = No)
Question Q24_5: Why do you not want to use such an information service?



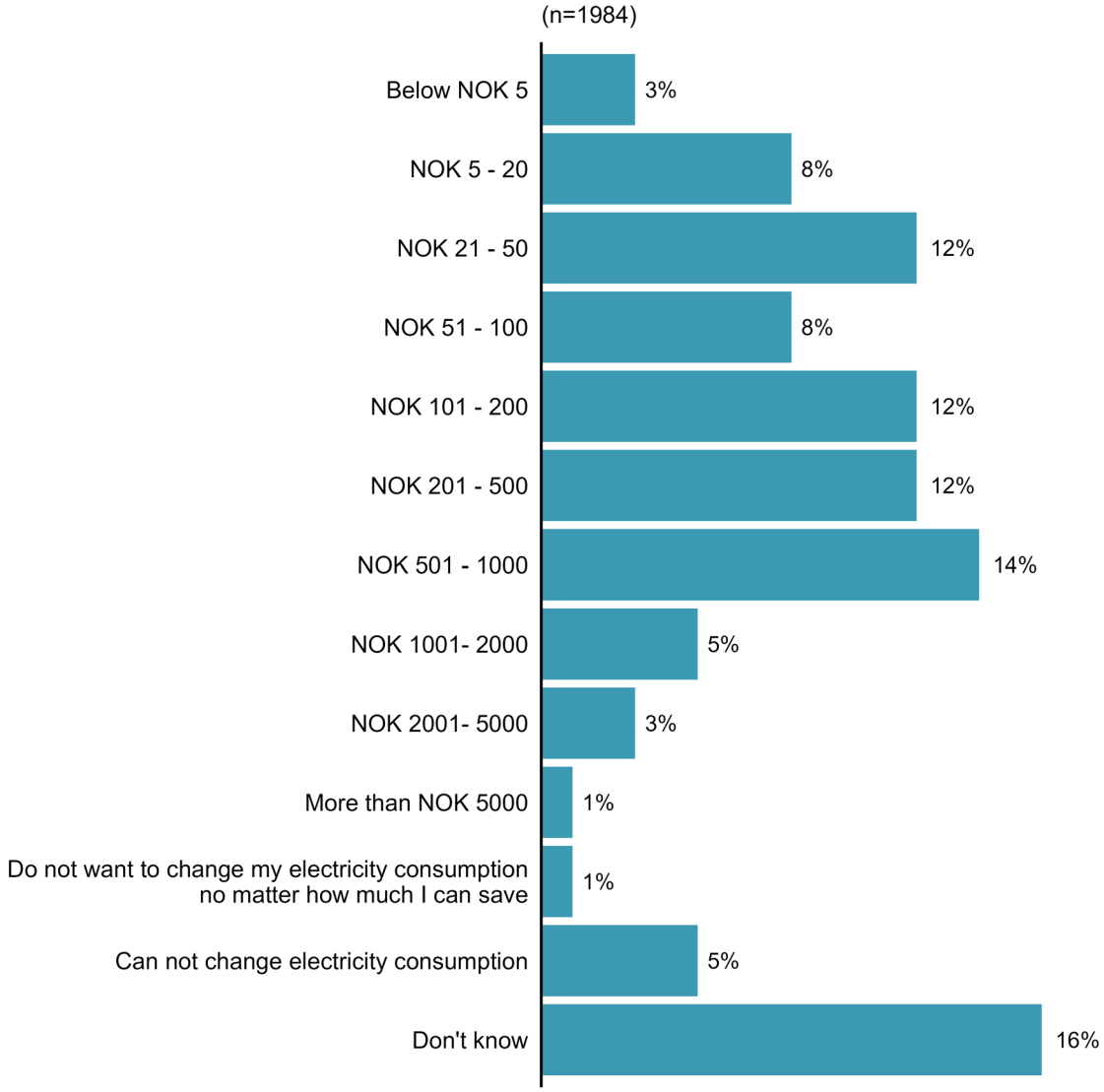
(IF Question Q20_11 = No answer, for all following questions)
Question Q25_a: Imagine that there is a high electricity price some weekdays (Monday - Friday) in winter. How much money do you have to save on the electricity bill to reduce electricity consumption a few hours in the middle of the day with a high electricity price… A few hours 1 weekday 1 winter month when the electricity price is high



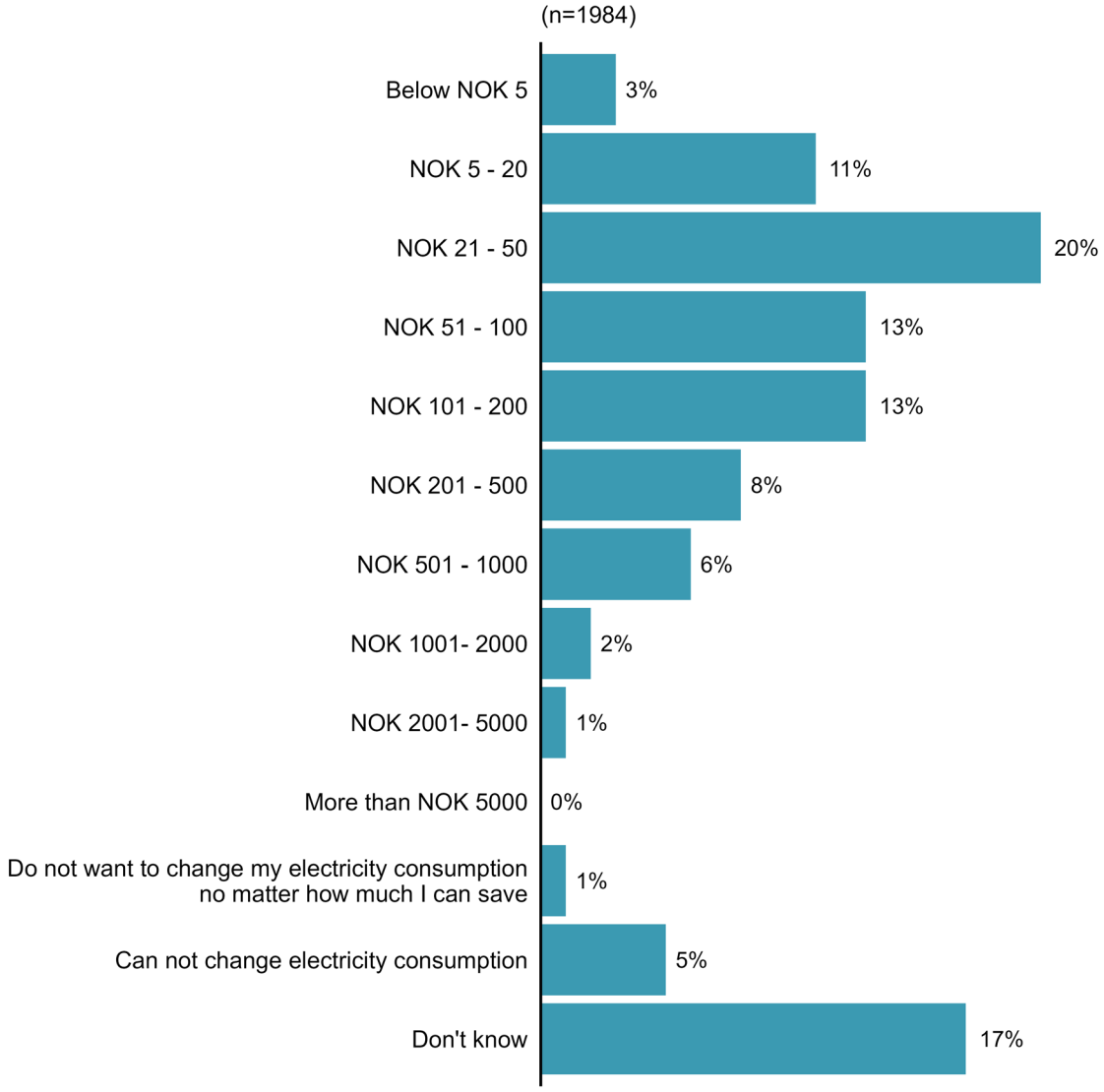
Question Q25_b: Imagine that there is a high electricity price some weekdays (Monday - Friday) in winter. How much money do you have to save on the electricity bill to reduce electricity consumption a few hours in the middle of the day with a high electricity price… A few hours 5 weekdays 1 winter month when the electricity price is high



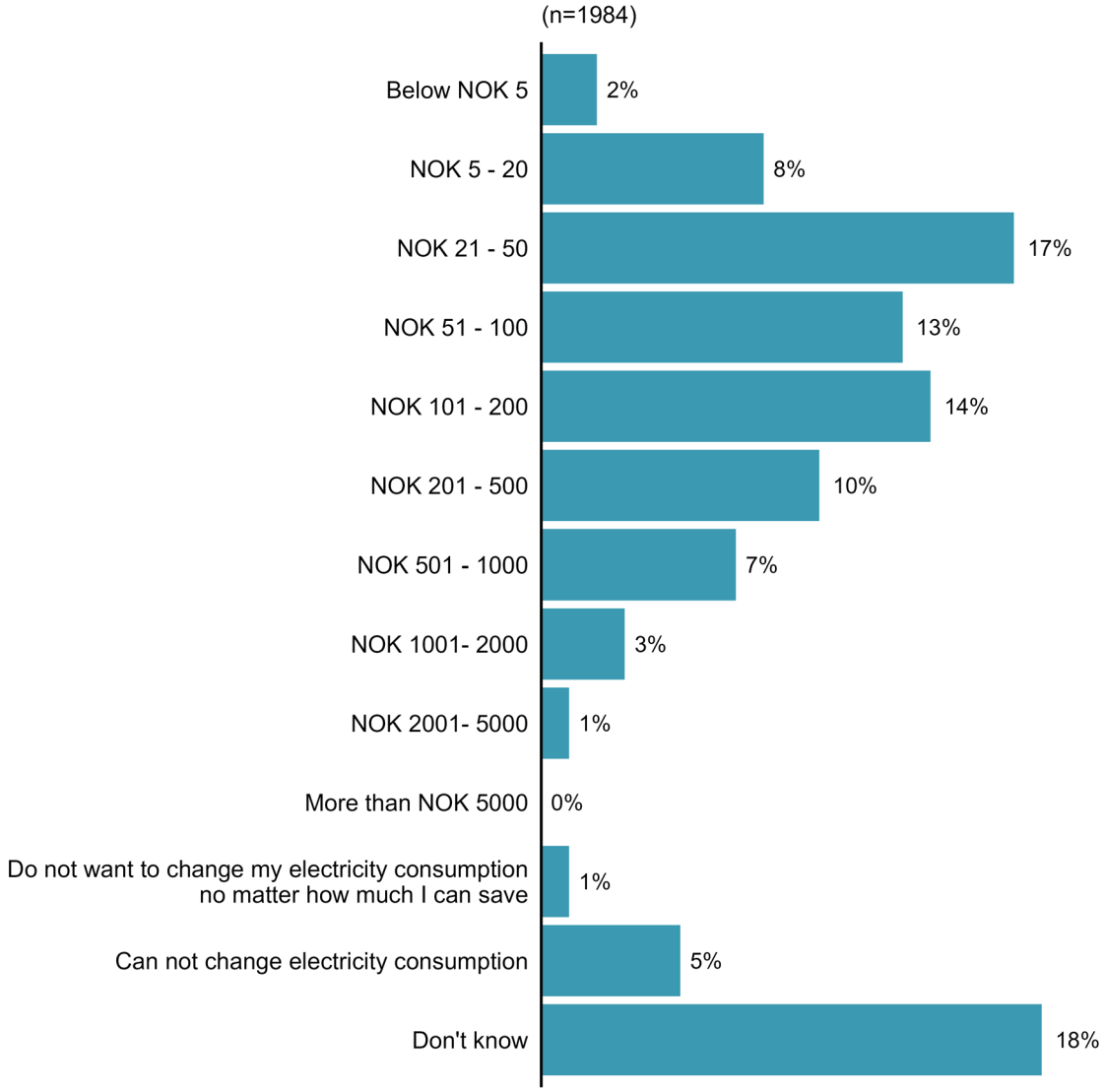
Question Q25_c: Imagine that there is a high electricity price some weekdays (Monday - Friday) in winter. How much money do you have to save on the electricity bill to reduce electricity consumption a few hours in the middle of the day with a high electricity price… A few hours all weekdays 1 winter month when the electricity price is high



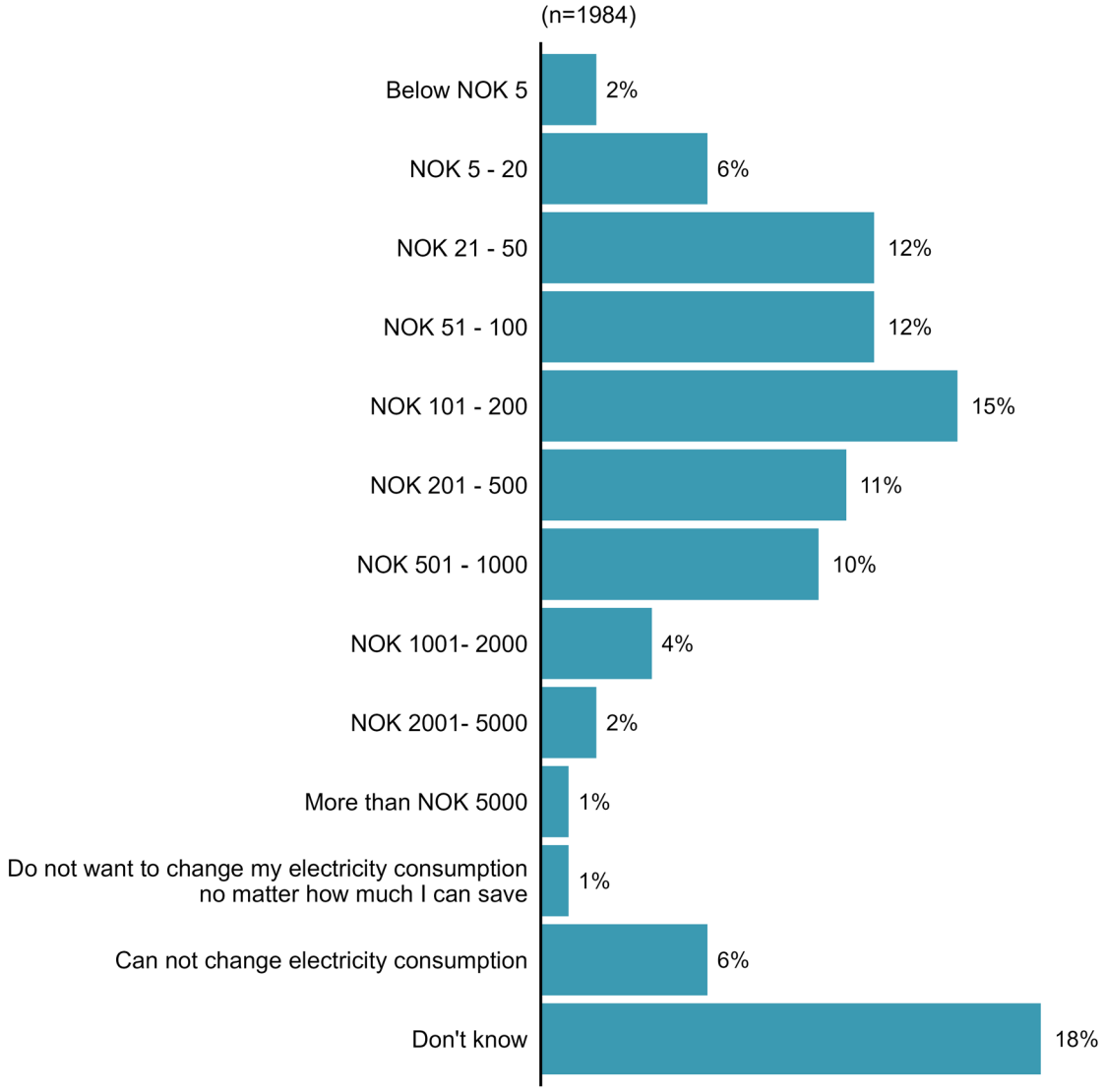
Question Q26_a: Imagine that there is a high electricity price for 2, 4, or 8 consecutive hours one weekday (Monday - Friday) in winter. How much money do you have to save on the electricity bill to reduce electricity consumption during these hours in the middle of the day with a high electricity price… 2 consecutive hours 1 weekday 1 winter month when the electricity price is high



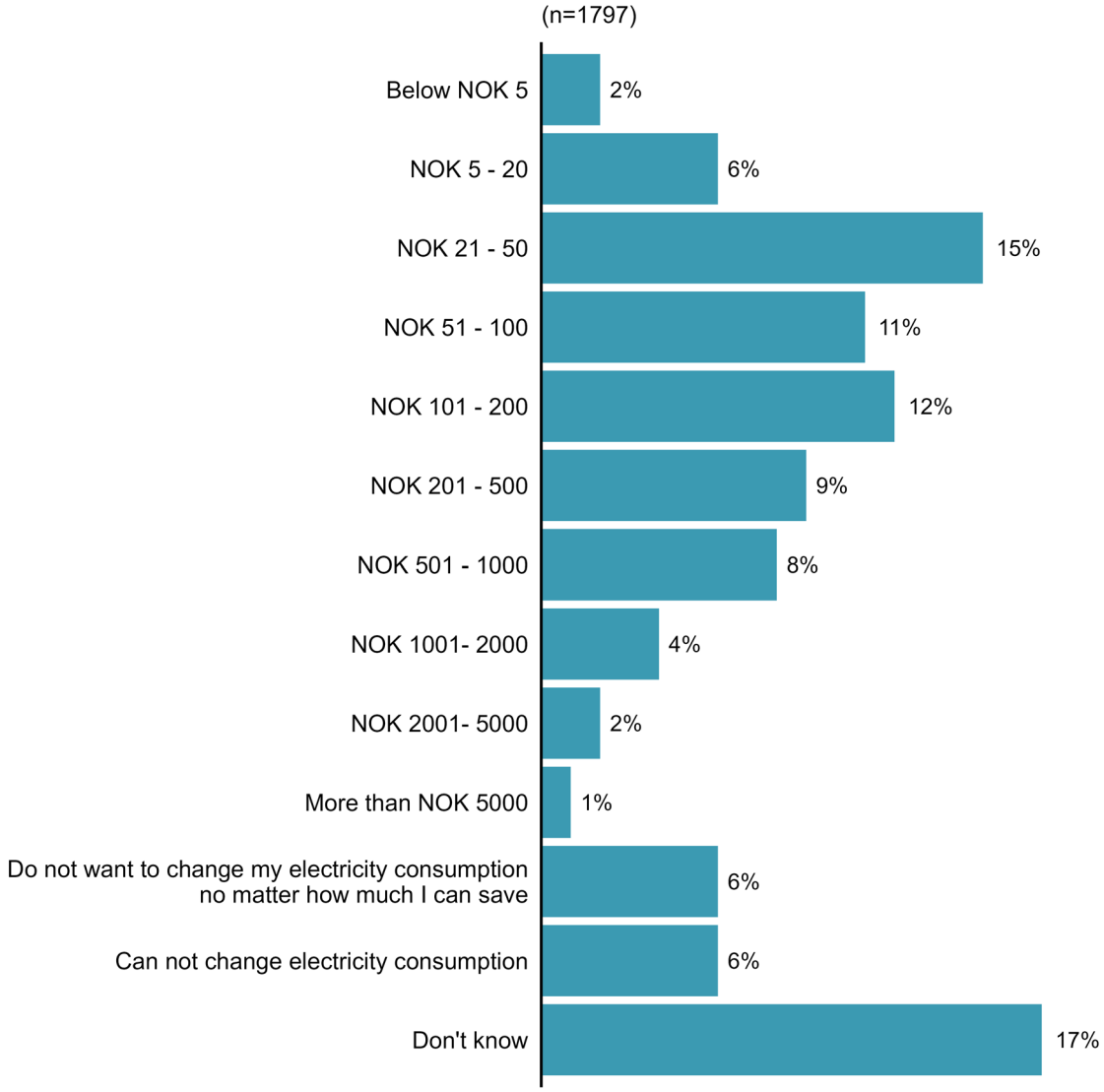
Question Q26_b: Imagine that there is a high electricity price for 2, 4, or 8 consecutive hours one weekday (Monday - Friday) in winter. How much money do you have to save on the electricity bill to reduce electricity consumption during these hours in the middle of the day with a high electricity price… 4 consecutive hours 1 weekday 1 winter month when the electricity price is high



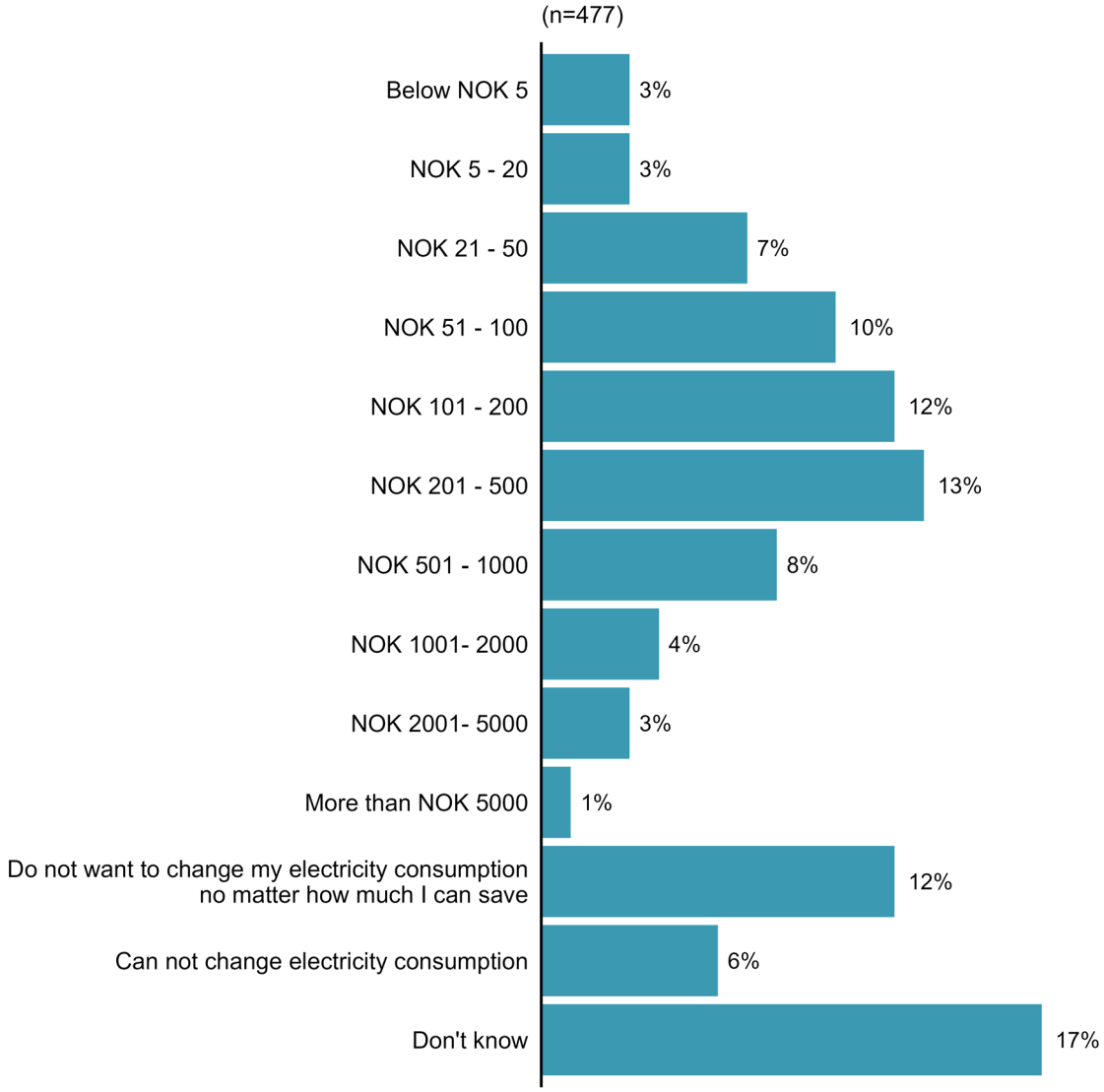
Question Q26_c: Imagine that there is a high electricity price for 2, 4, or 8 consecutive hours one weekday (Monday - Friday) in winter. How much money do you have to save on the electricity bill to reduce electricity consumption during these hours in the middle of the day with a high electricity price… 8 consecutive hours 1 weekday 1 winter month when the electricity price is high



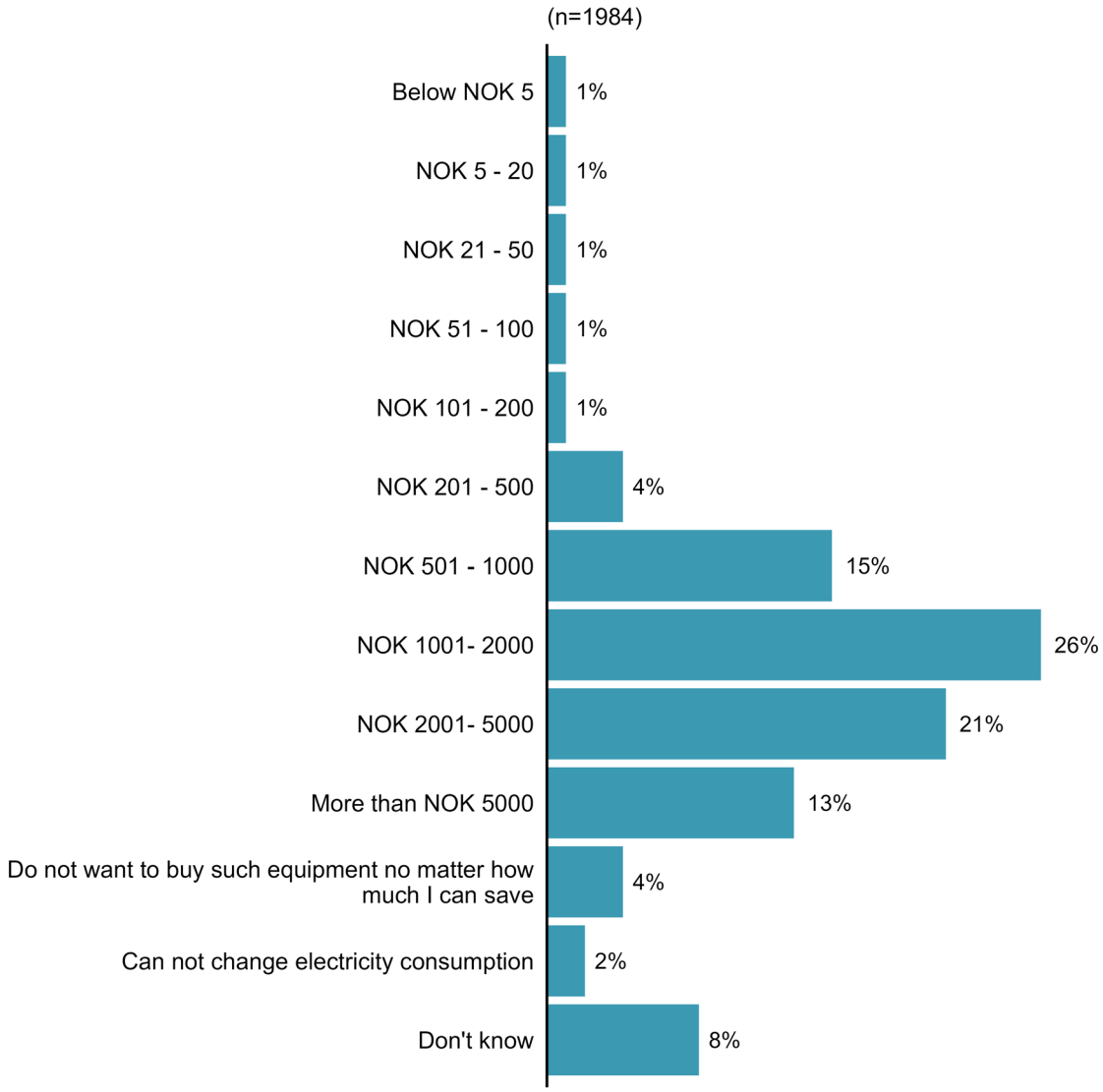
(IF Question Q6_1 OR Q6_2 OR Q6_3 OR Q6_4 = Yes)
Question Q27: How much money do you need to save on your electricity bill to reduce your electricity consumption with the following measures? Reduce indoor temperature by 3 degrees between kl. 16 and 20 on 1 weekday in the winter while you are at home



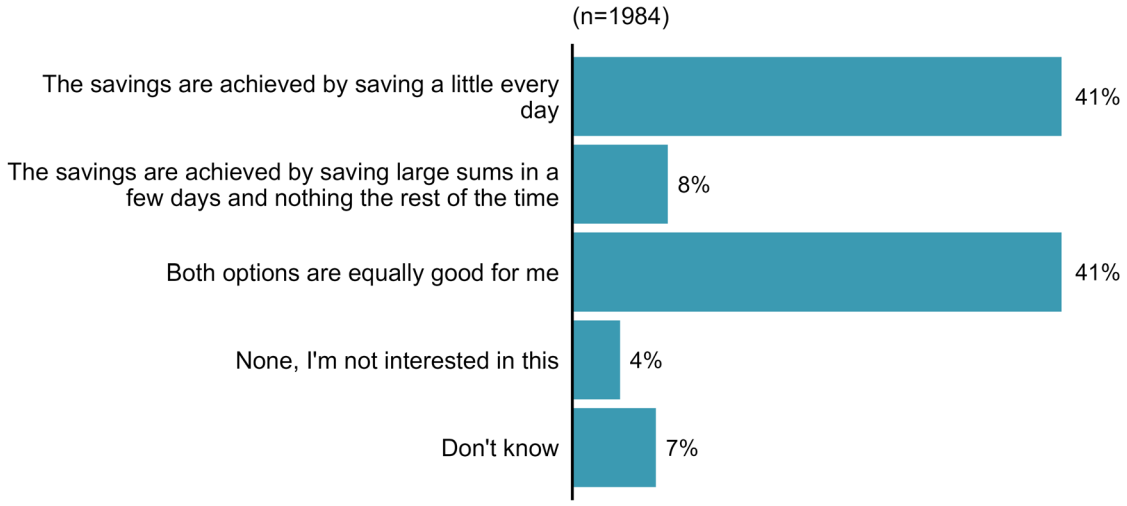
(IF Question Q8 = Yes)
Question Q28: How much money do you need to save on your electricity bill to reduce your electricity consumption with the following measures? Do not charge the electric car at home, so you have to stop 1 hour for charging on the next long car trip



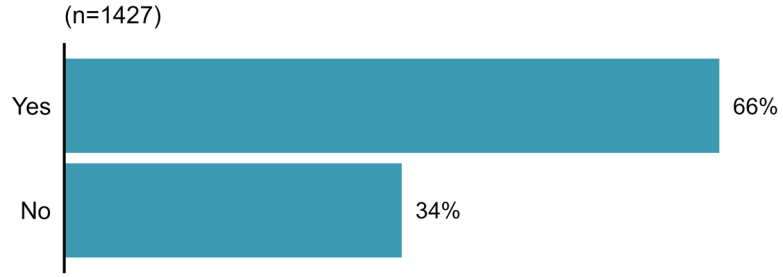
Question Q29: Imagine that you buy smart equipment that will reduce your electricity bill by automatically shifting parts of your consumption away from hours with high electricity prices, and you notice nothing. How much do you have to save per year to buy such smart equipment that costs NOK 5,000?



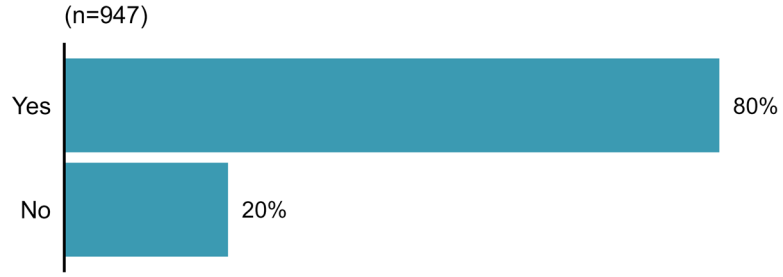
Question Q30: Imagine you could reduce your electricity bill by reducing your electricity consumption during hours of high electricity prices in winter. If you could choose which of the following options would you prefer?



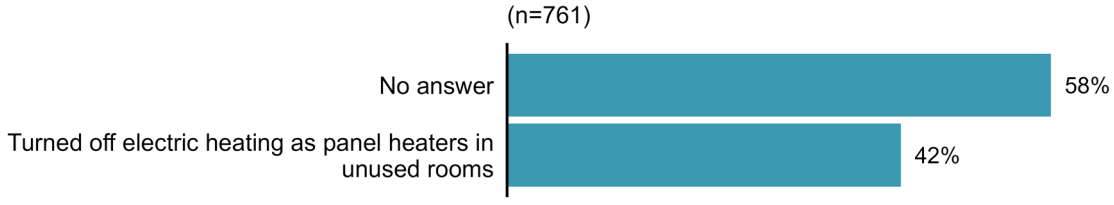
(IF part of a treatment group / not control group, for all following questions)
Question Q31: Your household participated in Statnett's price experiment during the winter months. We are interested in how you experienced the experiment and have therefore some questions. Did you actively follow the price signals given during the experiment?



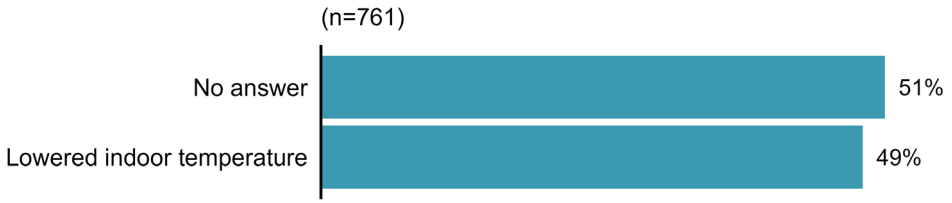
(IF Q31 = Yes)
Question Q32: Did you undertake measures in your household to reduce or shift electricity consumption during hours with high prices?



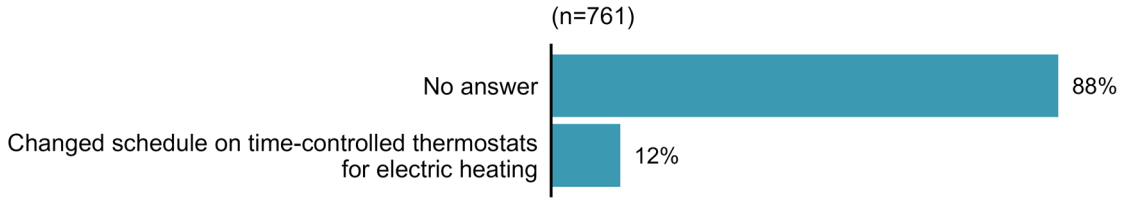
(IF Q32 = Yes)
Question Q33_1: What measures did you undertake? Turned off electric heating as panel heaters in unused rooms



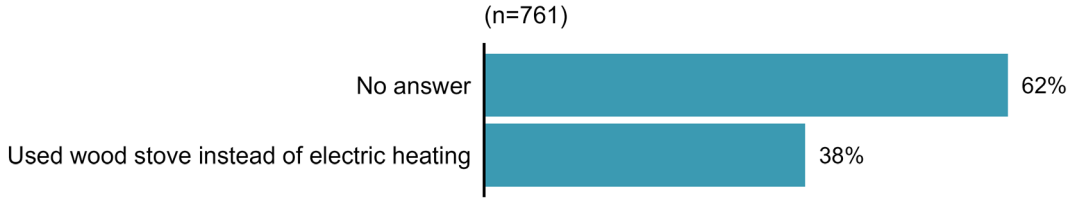
(IF Q32 = Yes)
Question Q33_2: What measures did you undertake? Lowered indoor temperature



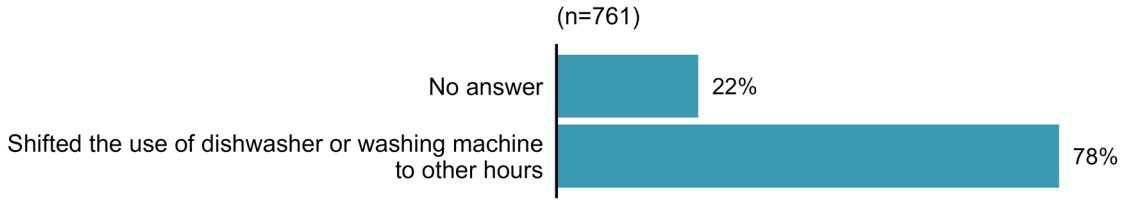
(IF Q32 = Yes)
Question Q33_3: What measures did you undertake? Changed schedule on time-controlled thermostats for electric heating



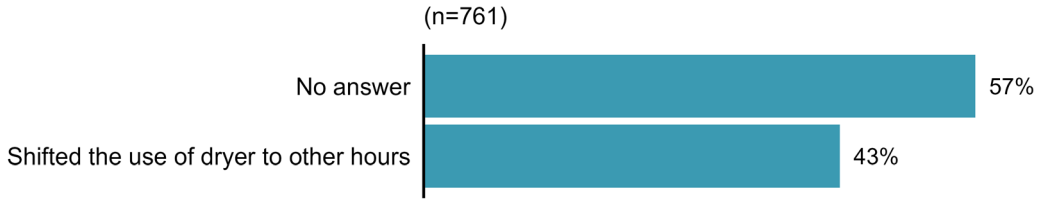
(IF Q32 = Yes)
Question Q33_4: What measures did you undertake? Used wood stove instead of electric heating



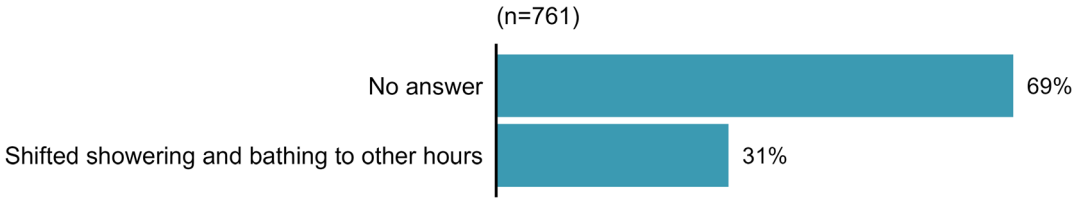
(IF Q32 = Yes)
Question Q33_5: What measures did you undertake? Shifted the use of dishwasher or washing machine to other hours



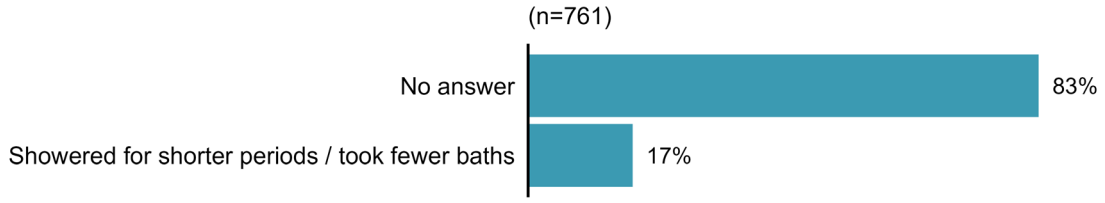
(IF Q32 = Yes)
Question Q33_6: What measures did you undertake? Shifted the use of dryer to other hours



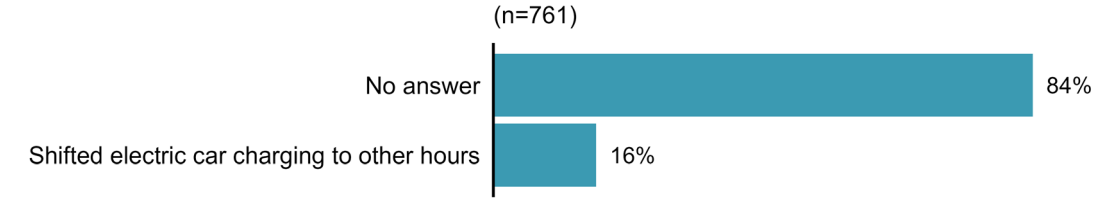
(IF Q32 = Yes)
Question Q33_7: What measures did you undertake? Shifted showering and bathing to other hours



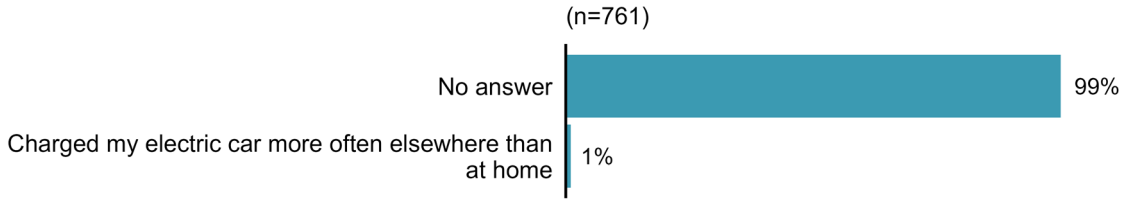
(IF Q32 = Yes)
Question Q33_8: What measures did you undertake? Showered for shorter periods / took fewer baths



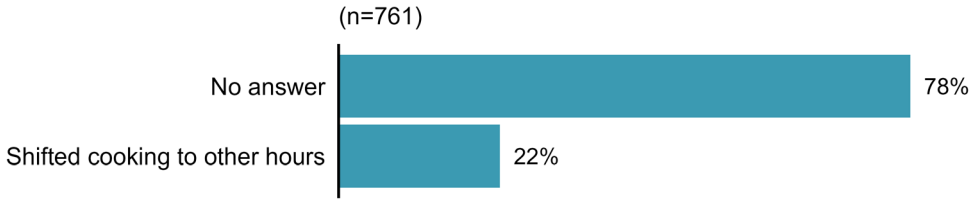
(IF Q32 = Yes)
Question Q33_9: What measures did you undertake? Shifted electric car charging to other hours



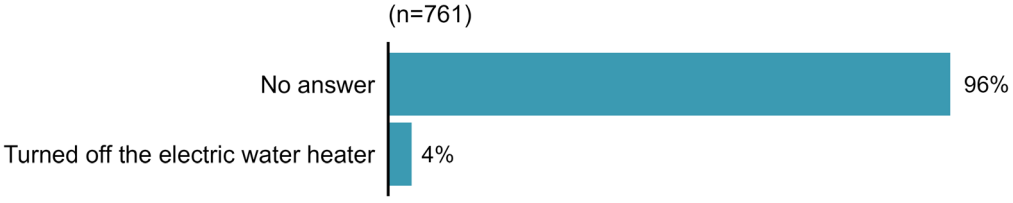
(IF Q32 = Yes)
Question Q33_10: What measures did you undertake? Charged my electric car more often elsewhere than at home



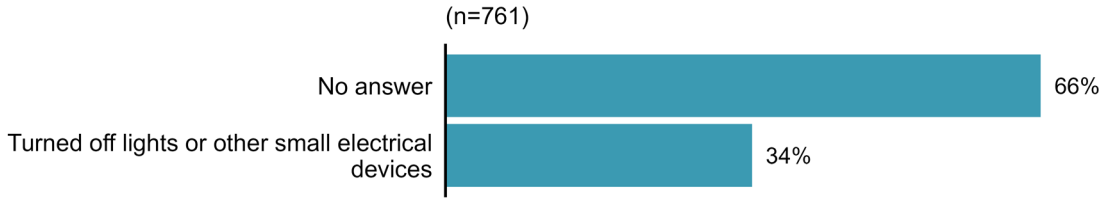
(IF Q32 = Yes)
Question Q33_11: What measures did you undertake? Shifted cooking to other hours



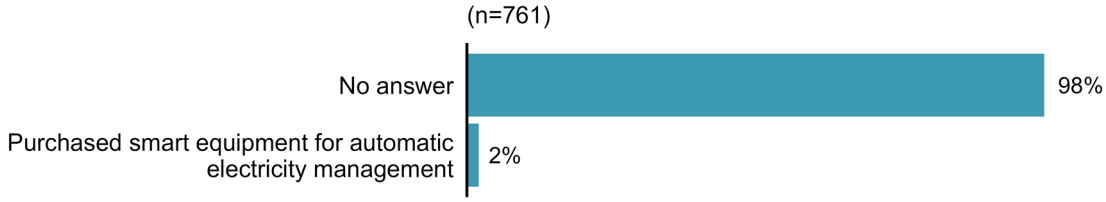
(IF Q32 = Yes)
Question Q33_12: What measures did you undertake? Turned off the electric water heater



(IF Q32 = Yes)
Question Q33_13: What measures did you undertake? Turned off lights or other small electrical devices



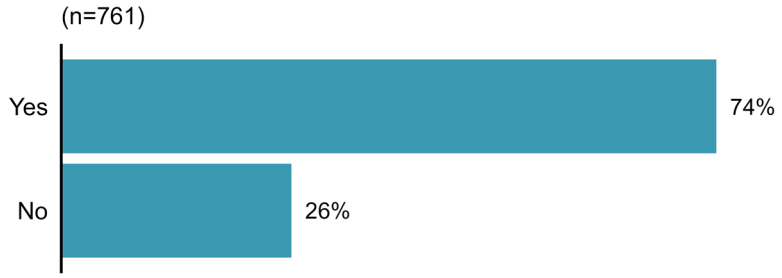
(IF Q32 = Yes)
Question Q33_14: What measures did you undertake? Purchased smart equipment for automatic electricity management



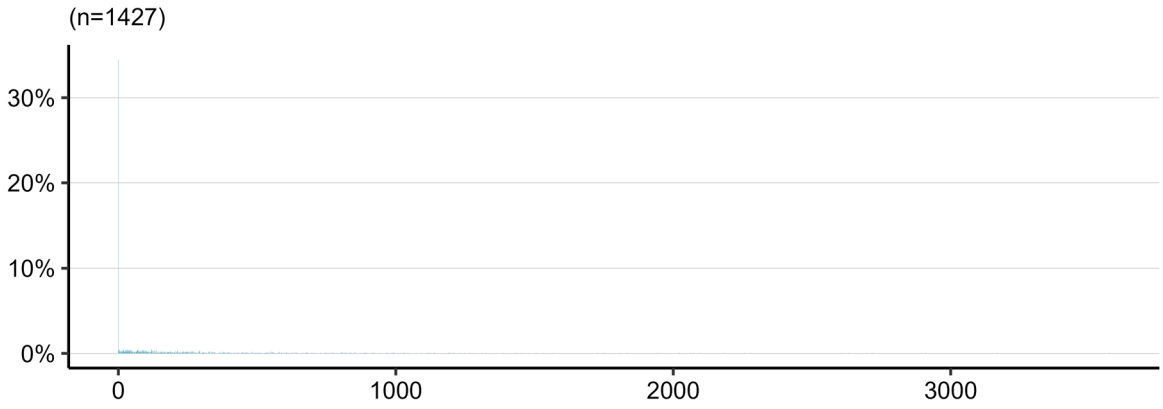
(IF Q32 = Yes)
Question Q33_15: What measures did you undertake? Other:



(IF Q32 = Yes)
Question Q34: You earned money during the price experiment thanks to your measures. Do you think it was worth undertaking the measures?



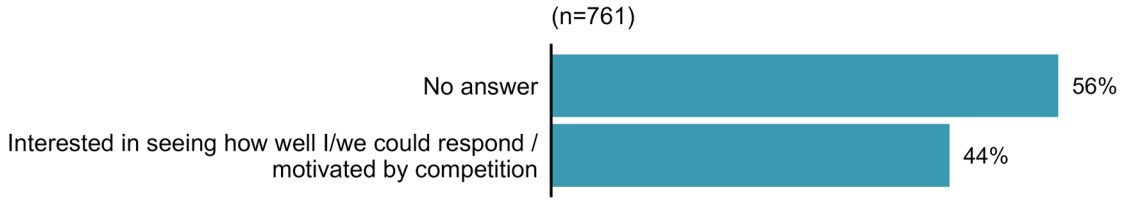
Question Q34_gevinst: Reward (Price group)



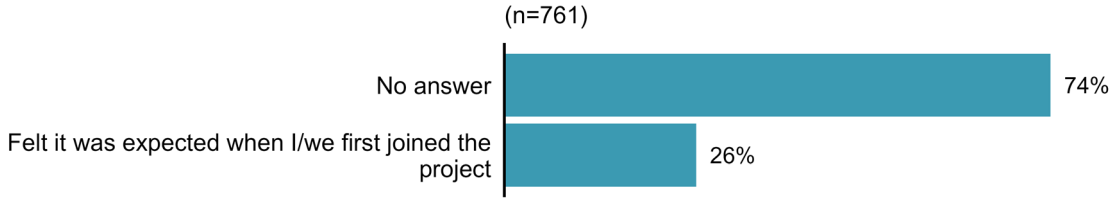
(IF Q32 = Yes)
Question Q35_1: Why did you undertake the measures? Opportunity to save/earn money



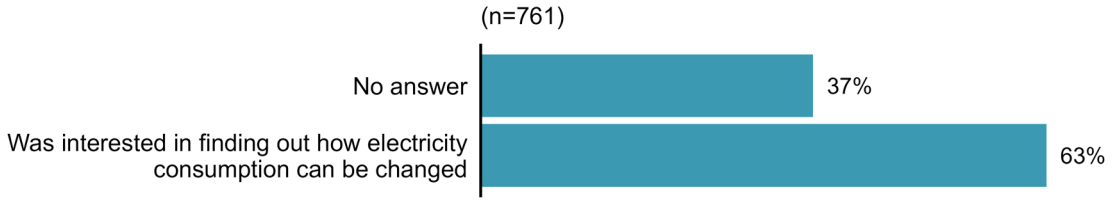
(IF Q32 = Yes)
Question Q35_2: Why did you undertake the measures? Interested in seeing how well I/we could respond / motivated by competition



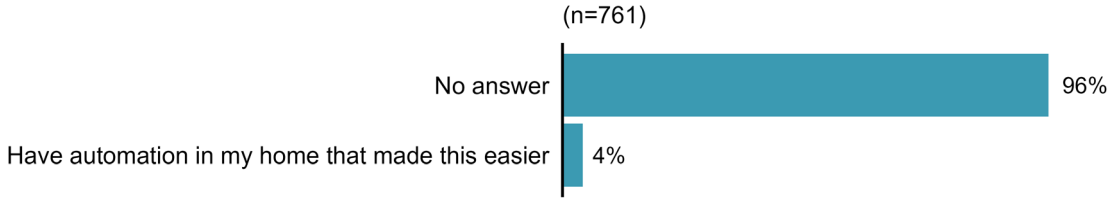
(IF Q32 = Yes)
Question Q35_3: Why did you undertake the measures? Felt it was expected when I/we first joined the project



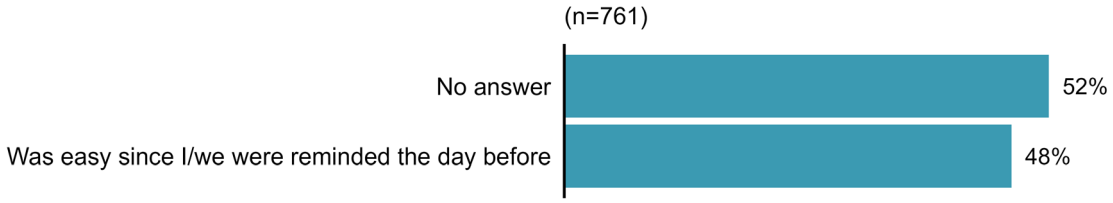
(IF Q32 = Yes)
Question Q35_4: Why did you undertake the measures? Was interested in finding out how electricity consumption can be changed



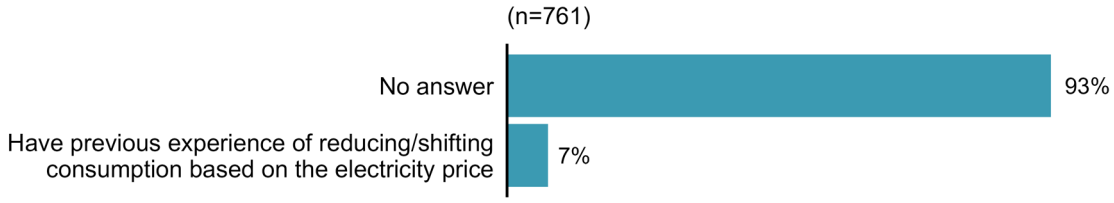
(IF Q32 = Yes)
Question Q35_5: Why did you undertake the measures? Have automation in my home that made this easier



(IF Q32 = Yes)
Question Q35_6: Why did you undertake the measures? Was easy since I/we were reminded the day before



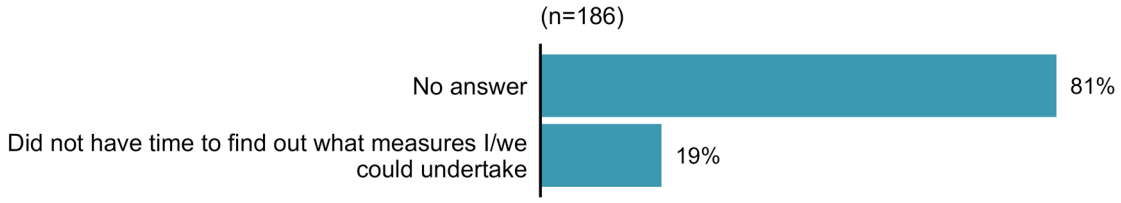
(IF Q32 = Yes)
Question Q35_7: Why did you undertake the measures? Have previous experience of reducing/shifting consumption based on the electricity price



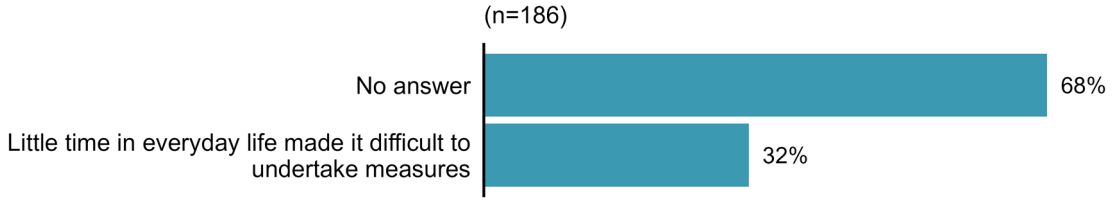
(IF Q32 = Yes)
Question Q35_8: Why did you undertake the measures? Other:



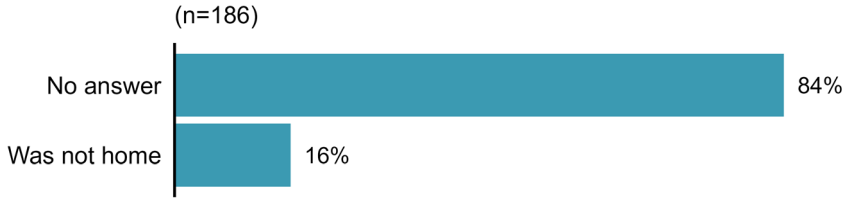
(IF Q32 = No)
Question Q36_1: Why did you not undertake such measures? Did not have time to find out what measures I/we could undertake



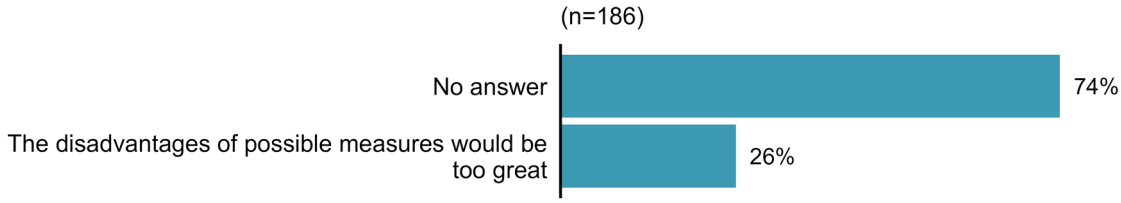
(IF Q32 = No)
Question Q36_2: Why did you not undertake such measures? Little time in everyday life made it difficult to undertake measures



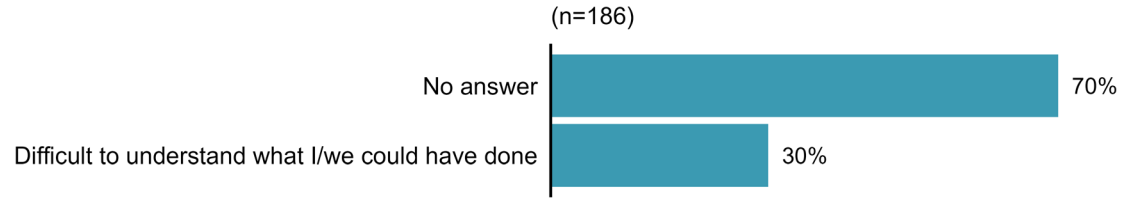
(IF Q32 = No)
Question Q36_3: Why did you not undertake such measures? Was not home



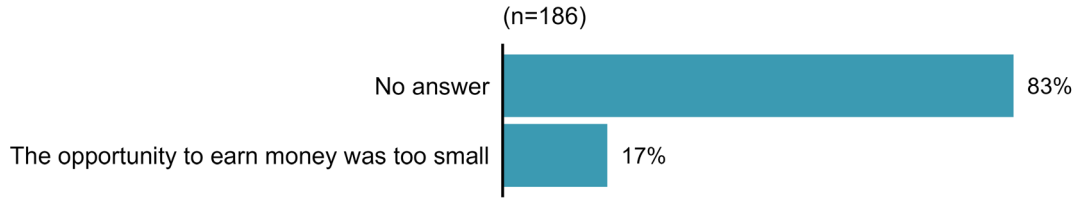
(IF Q32 = No)
Question Q36_4: Why did you not undertake such measures? The disadvantages of possible measures would be too great



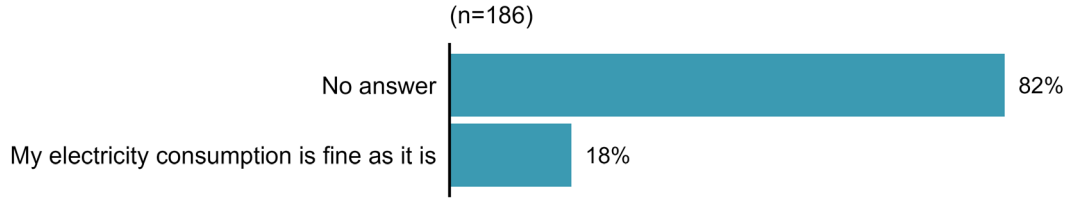
(IF Q32 = No)
Question Q36_5: Why did you not undertake such measures? Difficult to understand what I/we could have done



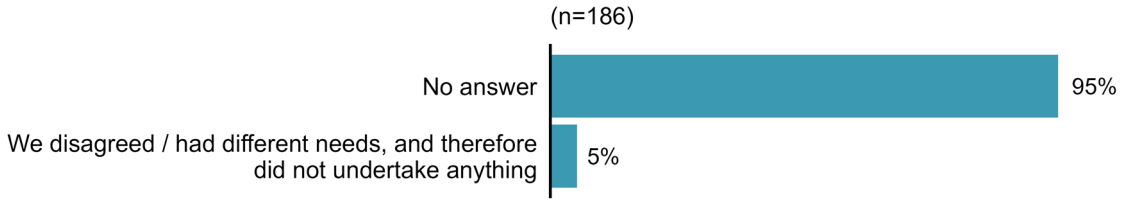
(IF Q32 = No)
Question Q36_6: Why did you not undertake such measures? The opportunity to earn money was too small



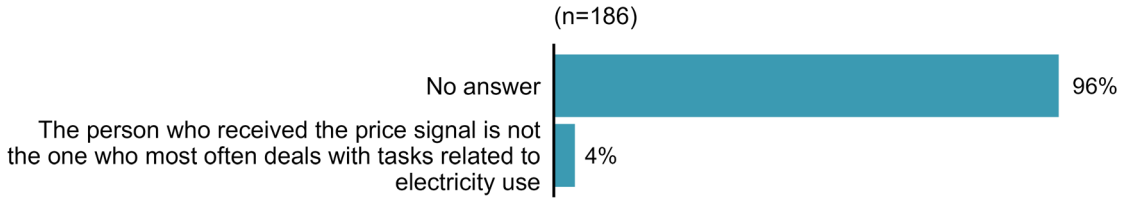
(IF Q32 = No)
Question Q36_7: Why did you not undertake such measures? My electricity consumption is fine as it is



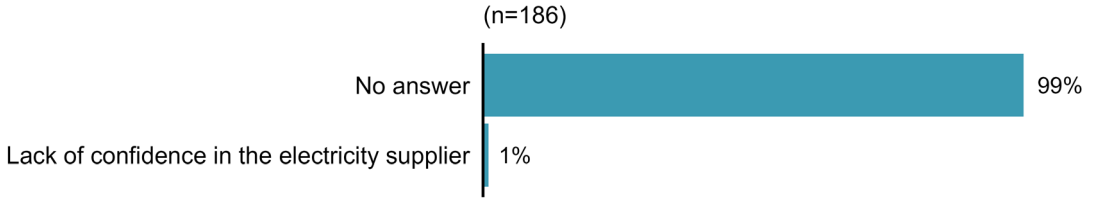
(IF Q32 = No)
Question Q36_8: Why did you not undertake such measures? We disagreed / had different needs, and therefore did not undertake anything



(IF Q32 = No)
Question Q36_9: Why did you not undertake such measures? The person who received the price signal is not the one who most often deals with tasks related to electricity use



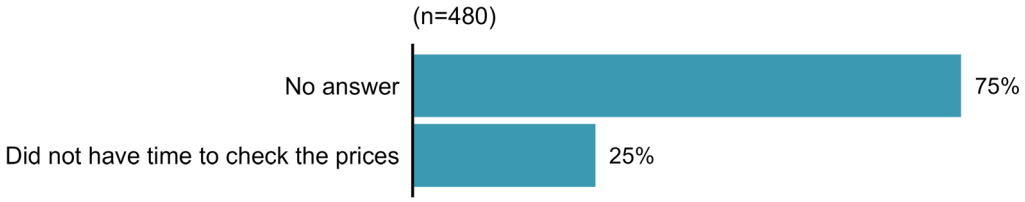
(IF Q32 = No)
Question Q36_10: Why did you not undertake such measures? Lack of confidence in the electricity supplier



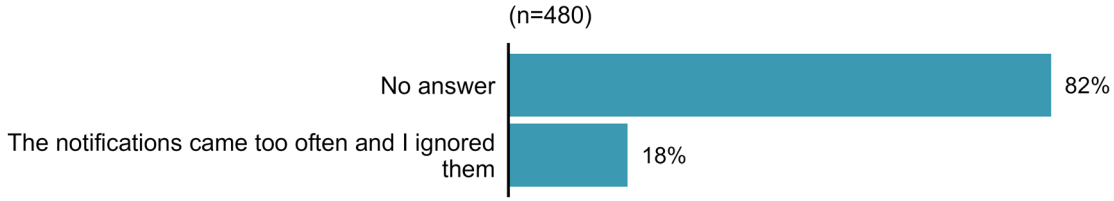
(IF Q32 = No)
Question Q36_11: Why did you not undertake such measures? Other:



(IF Q31 = No)
Question Q37_1: Why did you not actively follow the price signals given during the experiment? Did not have time to check the prices



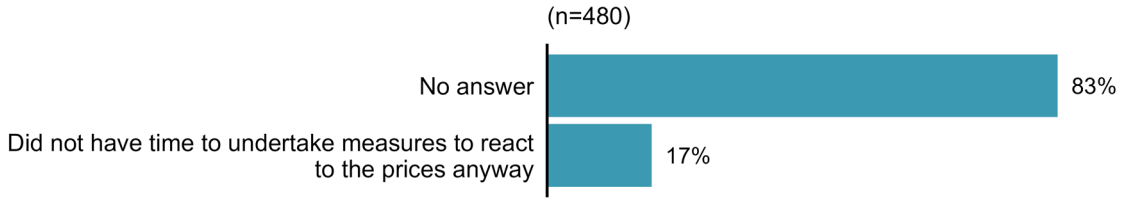
(IF Q31 = No)
Question Q37_2: Why did you not actively follow the price signals given during the experiment? The notifications came too often and I ignored them



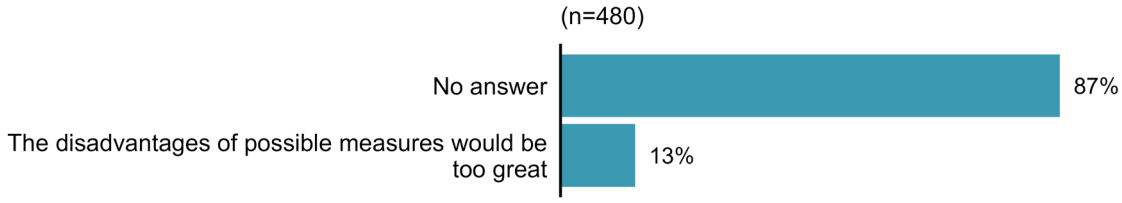
(IF Q31 = No)
Question Q37_3: Why did you not actively follow the price signals given during the experiment? Was not home



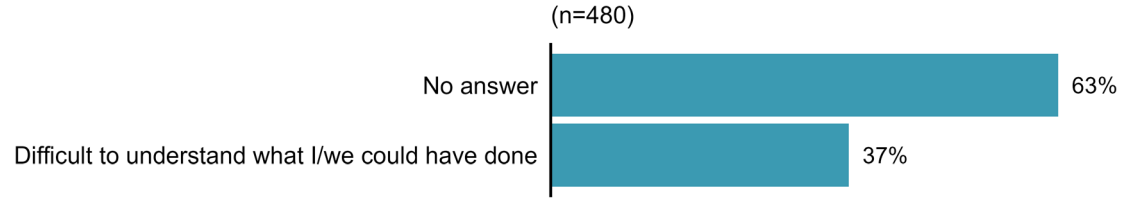
(IF Q31 = No)
Question Q37_4: Why did you not actively follow the price signals given during the experiment? Did not have time to undertake measures to react to the prices anyway



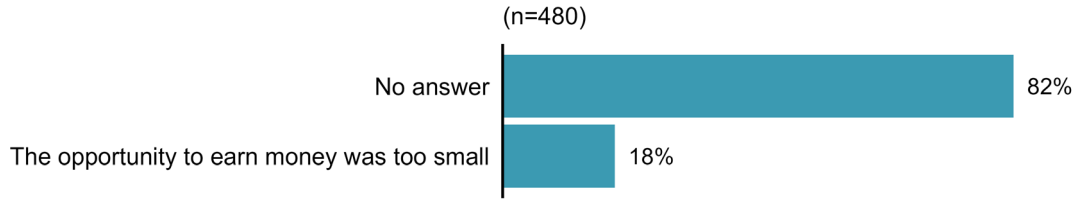
(IF Q31 = No)
Question Q37_5: Why did you not actively follow the price signals given during the experiment? The disadvantages of possible measures would be too great



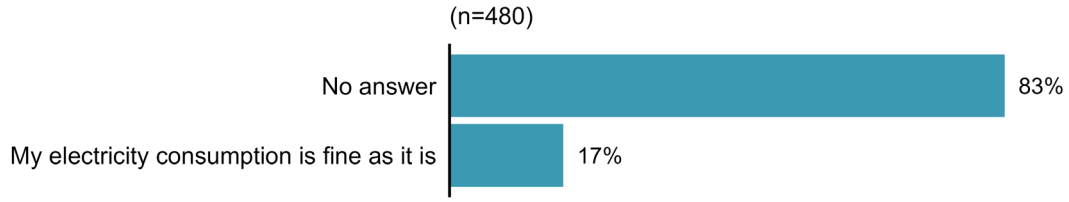
(IF Q31 = No)
Question Q37_6: Why did you not actively follow the price signals given during the experiment? Difficult to understand what I/we could have done



(IF Q31 = No)
Question Q37_7: Why did you not actively follow the price signals given during the experiment? The opportunity to earn money was too small



(IF Q31 = No)
Question Q37_8: Why did you not actively follow the price signals given during the experiment? My electricity consumption is fine as it is



(IF Q31 = No)
Question Q37_9: Why did you not actively follow the price signals given during the experiment? Other:



Question Q38: Is it okay for you to be contacted by researchers from the University of Oslo and the Norwegian University of Science and Technology to conduct an in-depth interview with you about power consumption? Households that conduct interviews are given a gift card of NOK 500.

